

CODING CHALLENGE

EMPLOYEE_DETAILS_ANALYSIS

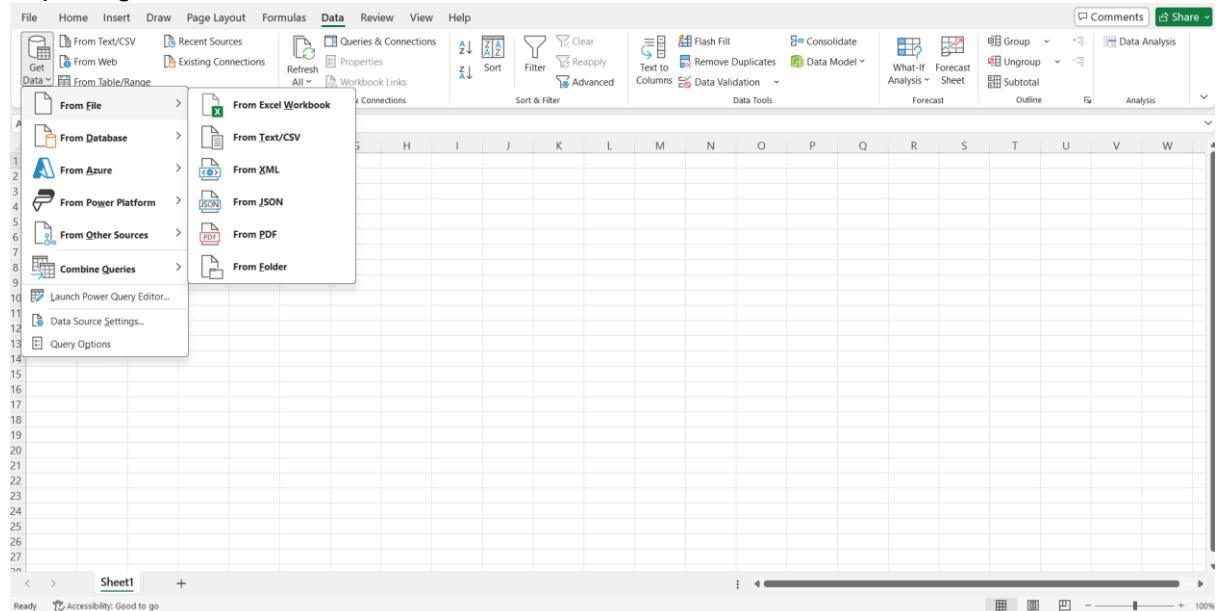
OVERVIEW:

The dataset contains employee-related information in an organization such as name, age, department, and salary. It includes both categorical and numerical data, making it suitable for analysis.

DAY1-Observation:

- The Employee Dataset consists of details of the employees, It has 9 columns and 151 rows.
- The Employee Dataset has 9 columns in which (Name, Department, Country, Gender) are categorical data and columns such as (Employee id, Salary, Date of Joining, Performance Rating) are Numerical Data.
- The Employee dataset contains some missing values and inconsistent data, which need to be cleaned before analysis.
- The Employee dataset can be used to understand employee distribution, salary patterns, and departmental insights.

Importing file into Excel:



File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query

Table Name: Sheet1

Properties: Summarize with PivotTable, Remove Duplicates, Convert to Range, Reshape Table

Tools: Insert Slicer, Export, Refresh, Open in Browser, Unlink

External Table Data

Table Design: Header Row, Total Row, Banded Rows, First Column, Last Column, Banded Columns, Filter Button

Table Styles

Employee ID	Name	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email
EMP1000		Finance		N/A		2	Unknown	invalid_email
EMP1001	Eve Adams	HR	Germany	54255	42015	1	Other	user106@example.com
EMP1002	Alice Brown	Finance	Germany	63184	42022	2		user176@example.com
EMP1003	John Doe	Sales	USA	94406	42029	1	Male	user168@example.com
EMP1004	Jane Smith	Sales	Germany	64120	42036	1	Male	user49@example.com
EMP1005	Alice Brown	IT	India	78421	42043	2	Female	user24@example.com
EMP1006	Sam Wilson	Sales		67721	42050		Male	user168@example.com
EMP1007	Chris Lee	Marketing		35364	42057	2	Other	user26@example.com
EMP1008	Alice Brown	IT	Germany	97881	42064		Female	user25@example.com
EMP1009	Chris Lee	HR	UK	91846	42071		Male	user153@example.com
EMP1010	Sam Wilson	Finance	UK	64247	42078	1	Female	user18@example.com
EMP1011	Sam Wilson	IT	USA	91180	42085	5	Male	user1@example.com
EMP1012	Bob Johnson	Sales		96418	42092	5	Male	user106@example.com
EMP1013	Sam Wilson		India	62158	42099	4		user180@example.com
EMP1014	Jane Smith	HR		59894	42106	2	Male	user184@example.com
EMP1015		IT	India	92443	42113	3	Other	user3@example.com
EMP1016	Bob Johnson		Germany	90271	42120			user64@example.com
EMP1017	Jane Smith	Finance	Australia	41341	42127	1	Other	user38@example.com
EMP1018	Alice Brown	Finance	UK	73602	42134			user96@example.com
EMP1019	Eve Adams		India	67531	42141	5	Female	user151@example.com
EMP1020	Jane Smith	HR	USA	N/A	42148	2	Other	user200@example.com
EMP1021	Jane Smith	HR		98922	42155	4		user77@example.com
EMP1022	Eve Adams	Finance	India	38333	42162	5		user128@example.com
EMP1023	Alice Brown	Marketing	UK	36515	42169	3	Male	user149@example.com
EMP1024	Sam Wilson	Finance		91005	42176	1	Other	user122@example.com
EMP1025	Chris Lee			52272	42183	3	Female	user41@example.com

DAY2- Data Entry & Formatting:

- A new column named "Employee First Name" and an extra row were added for practice, and later both were removed using the delete options in the Home tab.

File Home Insert Draw Page Layout Formulas Data Review View Help

Clipboard: Cut, Copy, Paste, Format Painter

Font: Calibri, Font Size, Bold, Italic, Underline, Paragraph, Text Color, Fill Color, Background Color

Alignment: Wrap Text, Merge & Center, Center, Left, Right, Indent, Decrease Indent, Increase Indent

Number: General, Number, Currency, Percentage, Date, Time, Text, Fraction, Scientific, Custom

Conditional Formatting: Highlight Cells, Top/Bottom, Data Bars, Color Scales, Icon Sets

Format as Table: Table Styles

Cell Styles: Cell Styles

Insert: Rows, Columns, Tables, Charts, Links, Comments, Hyperlinks

Delete: Rows, Columns, Cells, Comments, Hyperlinks

Format: Cells, Conditional Formatting, Table Styles, Cell Styles

AutoSum, Fill, Clear

Editing: Sort & Filter, Find & Select, Add-ins

Employee ID	Name	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email	Employee first name
EMP1000		Finance		N/A		2	Unknown	invalid_email	
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	1	Other	user106@example.com	Eve
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2		user176@example.com	Ali
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com	John
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com	Jan
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com	Ali
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015		Male	user168@example.com	Sam
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015	2	Other	user26@example.com	Chr
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015		Female	user25@example.com	Ali
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015		Male	user153@example.com	Chr
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com	Sam
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com	Sam
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015	5	Male	user106@example.com	Bob
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015	4		user180@example.com	Sam
EMP1121	Chris Lee	Sales	USA	₹ 45,676.00	30-04-2017			user132@example.com	Chr
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015	2	Male	user184@example.com	Jan
EMP1015		IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com	
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015			user64@example.com	Bob
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com	Jan
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015			user96@example.com	Ali
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com	Eve
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015	2	Other	user200@example.com	Jan
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015	4		user77@example.com	Jan
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5		user128@example.com	Eve
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com	Ali
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015	1	Other	user122@example.com	Sam

- The dataset was formatted using table design, and the header row was highlighted by increasing the font size and applying bold formatting.
- The Salary column was formatted into currency type, and the Date column was converted into short date format for better readability.

Proper alignment was applied, where text values were aligned to the left and numerical values were aligned to the right

Employee ID	Name	Department	Country	Salary	Date of Join	Performance Rating	Gender	Email
EMP1000	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	2	Unknown	invalid_email
EMP1001	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	1	Other	user106@example.com
EMP1002	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	2	Male	user176@example.com
EMP1003	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com
EMP1004	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com
EMP1005	Sam Wilson	Sales	India	₹ 67,721.00	15-02-2015	2	Male	user168@example.com
EMP1006	Chris Lee	Marketing	UK	₹ 35,364.00	22-02-2015	2	Other	user26@example.com
EMP1007	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015	1	Female	user25@example.com
EMP1008	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015	1	Male	user153@example.com
EMP1009	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com
EMP1010	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com
EMP1011	Bob Johnson	Sales	India	₹ 96,418.00	29-03-2015	5	Male	user106@example.com
EMP1012	Sam Wilson	HR	India	₹ 62,158.00	05-04-2015	4	Male	user180@example.com
EMP1013	Jane Smith	HR	India	₹ 59,894.00	12-04-2015	2	Male	user184@example.com
EMP1014	Jane Smith	IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com
EMP1015	Bob Johnson	Finance	Germany	₹ 90,271.00	26-04-2015	1	Other	user64@example.com
EMP1016	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com
EMP1017	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015	1	Other	user96@example.com
EMP1018	Eve Adams	HR	India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com
EMP1019	Jane Smith	HR	USA	₹ 98,922.00	31-05-2015	2	Other	user200@example.com
EMP1020	Jane Smith	HR	USA	₹ 98,922.00	31-05-2015	2	Other	user77@example.com
EMP1021	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5	Male	user128@example.com
EMP1022	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com
EMP1023	Sam Wilson	Finance	India	₹ 91,005.00	21-06-2015	1	Other	user122@example.com
EMP1024	Chris Lee	Finance	India	₹ 52,272.00	28-06-2015	3	Other	user41@example.com

DAY3-Basic Excel Formulas (Part 1).

SUM(), AVERAGE(), MIN(), MAX(), COUNT()

Employee ID	Name	Department	Country	Salary	Date of Join	Performance Rating	Gender	Email
EMP1000	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	2	Unknown	invalid_email
EMP1001	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	1	Other	user106@example.com
EMP1002	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	2	Male	user176@example.com
EMP1003	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com
EMP1004	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com
EMP1005	Sam Wilson	Sales	India	₹ 67,721.00	15-02-2015	2	Male	user168@example.com
EMP1006	Chris Lee	Marketing	UK	₹ 35,364.00	22-02-2015	2	Other	user26@example.com
EMP1007	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015	1	Female	user25@example.com
EMP1008	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015	1	Male	user153@example.com
EMP1009	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com
EMP1010	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com
EMP1011	Bob Johnson	Sales	India	₹ 96,418.00	29-03-2015	5	Male	user106@example.com
EMP1012	Sam Wilson	HR	India	₹ 62,158.00	05-04-2015	4	Male	user180@example.com
EMP1013	Jane Smith	HR	India	₹ 59,894.00	12-04-2015	2	Male	user184@example.com
EMP1014	Jane Smith	IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com
EMP1015	Bob Johnson	Finance	Germany	₹ 90,271.00	26-04-2015	1	Other	user64@example.com
EMP1016	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com
EMP1017	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015	1	Other	user96@example.com
EMP1018	Eve Adams	HR	India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com
EMP1019	Jane Smith	HR	USA	₹ 98,922.00	31-05-2015	2	Other	user200@example.com
EMP1020	Jane Smith	HR	USA	₹ 98,922.00	31-05-2015	2	Other	user77@example.com
EMP1021	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5	Male	user128@example.com
EMP1022	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com
EMP1023	Sam Wilson	Finance	India	₹ 91,005.00	21-06-2015	1	Other	user122@example.com
EMP1024	Chris Lee	Finance	India	₹ 52,272.00	28-06-2015	3	Other	user41@example.com

Metric	Salary
SUM	₹ 94,11,892.00
AVERAGE	₹ 66,280.93
MIN	₹ 30,280.00
MAX	₹ 99,621.00
COUNT	142

Metric	PERFORMANCE RATING
SUM	359
AVERAGE	2.872
MIN	1
MAX	1
COUNT	125
COUNTBLANK	25

SUM function is used to calculate the total salary of the employees.

Employee ID Name Department Country Salary Date of Join Performance Ratio Gender Email																
EMP1000		Finance		N/A			2	Unknown	invalid_email							
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015		1	Other	user106@example.com							
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015		2		user176@example.com							
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015		1	Male	user168@example.com							
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015		1	Male	user49@example.com							
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015		2	Female	user24@example.com							
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015			Male	user168@example.com							
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015		2	Other	user26@example.com							
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015			Female	user25@example.com							
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015			Male	user153@example.com							
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015		1	Female	user18@example.com							
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015		5	Male	user1@example.com							
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015		5	Male	user106@example.com							
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015		4		user180@example.com							
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015		2	Male	user184@example.com							
EMP1015		IT	India	₹ 92,443.00	19-04-2015		3	Other	user3@example.com							
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015				user64@example.com							
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015		1	Other	user38@example.com							
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015				user96@example.com							
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015		5	Female	user151@example.com							
EMP1020	Jane Smith	HR	USA		24-05-2015		2	Other	user200@example.com							
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015		4		user77@example.com							
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015		5		user128@example.com							
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015		3	Male	user149@example.com							
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015		1	Other	user122@example.com							
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015		3		user41@example.com							

Average function is used to find the typical salaries of the employees

Employee ID Name Department Country Salary Date of Join Performance Ratio Gender Email																
EMP1000		Finance		N/A			2	Unknown	invalid_email							
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015		1	Other	user106@example.com							
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015		2		user176@example.com							
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015		1	Male	user168@example.com							
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015		1	Male	user49@example.com							
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015		2	Female	user24@example.com							
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015			Male	user168@example.com							
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015		2	Other	user26@example.com							
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015			Female	user25@example.com							
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015			Male	user153@example.com							
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015		1	Female	user18@example.com							
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015		5	Male	user1@example.com							
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015		5	Male	user106@example.com							
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015		4		user180@example.com							
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015		2	Male	user184@example.com							
EMP1015		IT	India	₹ 92,443.00	19-04-2015		3	Other	user3@example.com							
EMP1016	Bob Johnson	Finance	Germany	₹ 90,271.00	26-04-2015				user64@example.com							
EMP1017	Jane Smith		Australia	₹ 41,341.00	03-05-2015		1	Other	user38@example.com							
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015				user96@example.com							
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015		5	Female	user151@example.com							
EMP1020	Jane Smith	HR	USA		24-05-2015		2	Other	user200@example.com							
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015		4		user77@example.com							
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015		5		user128@example.com							
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015		3	Male	user149@example.com							
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015		1	Other	user122@example.com							
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015		3		user41@example.com							

MIN function is used to find the Minimum salary of the employees

File

Home

Insert

Draw

Page Layout

Formulas

Data

Review

View

Help

Cut

Copy

Paste

Format Painter

Clipboard

Calibri

11

A⁺

Wrap Text

Accounting

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

AutoSum

Fill

Clear

Sort & Filter

Find & Select

Add-ins

Editing

Add-ins

File tab

</

Max function is used to find the maximum salary of the employees

File

Home

Insert

Draw

Page Layout

Formulas

Data

Review

View

Help

Clipboard

Cut

Copy

Paste

Format Painter

Font

Alignment

Wrap Text

Merge & Center

General

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

AutoSum

Sort & Filter

Find & Select

Comments

Share

Add-ins

Clipboard

Cut

Copy

Paste

Format Painter

Font

Alignment

Wrap Text

Merge & Center

General

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

AutoSum

Sort & Filter

Find & Select

Comments

Share

Add-ins

Clipboard

Cut

Copy

Paste

Format Painter

Font

Alignment

Wrap Text

Merge & Center

General

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

AutoSum

Sort & Filter

Find & Select

Comments

Share

Add-ins

L10

</

Count value is used to find how many employee have salary value. the COUNT function returned a lower value, which shows that some cells contain non-numeric or invalid data instead of proper salary values.

Likewise these formulas are also used to calculate the total, minimum, maximum values of performance rating column.

File Home Insert Draw Page Layout Formulas Data Review View Help										Comments		Share	
Cut Copy Format Painter Clipboard Font Alignment Number Styles Wrap Text Merge & Center Conditional Formatting Format as Table Cell Styles Insert Delete Format										AutoSum Fill Clear		Sort & Filter Find & Select Add-ins	
L15													
=SUM(G:G)													
Employee ID	Name	Department	Country	Salary	Date of Join	Performance Rating	Gender	Email					
EMP1000		Finance		N/A			2 Unknown	invalid_email					
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015		1 Other	user106@example.com					
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015		2	user176@example.com					
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015		1 Male	user168@example.com					
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015		1 Male	user49@example.com					
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015		2 Female	user24@example.com					
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015		Male	user168@example.com					
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015		2 Other	user26@example.com					
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015		Female	user25@example.com					
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015		Male	user153@example.com					
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015		1 Female	user18@example.com					
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015		5 Male	user1@example.com					
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015		5 Male	user106@example.com					
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015		4	user180@example.com					
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015		2 Male	user184@example.com					
EMP1015		IT	India	₹ 92,443.00	19-04-2015		3 Other	user3@example.com					
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015			user64@example.com					
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015		1 Other	user38@example.com					
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015			user96@example.com					
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015		5 Female	user151@example.com					
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015		2 Other	user200@example.com					
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015		4	user77@example.com					
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015		5	user128@example.com					
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015		3 Male	user149@example.com					
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015		1 Other	user122@example.com					
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015		3	user41@example.com					

Metric	Salary
SUM	₹ 94,11,892.00
AVERAGE	₹ 66,280.93
MIN	₹ 30,280.00
MAX	₹ 99,621.00
COUNT	142

Metric	PERFORMANCE RATING
SUM	359
AVERAGE	2.872
MIN	1
MAX	5
COUNT	125
COUNTBLANK	25

File Home Insert Draw Page Layout Formulas Data Review View Help										Comments		Share	
Cut Copy Format Painter Clipboard Font Alignment Number Styles Wrap Text Merge & Center Conditional Formatting Format as Table Cell Styles Insert Delete Format										AutoSum Fill Clear		Sort & Filter Find & Select Add-ins	
L16													
=AVERAGE(G:G)													
Employee ID	Name	Department	Country	Salary	Date of Join	Performance Rating	Gender	Email					
EMP1000		Finance		N/A			2 Unknown	invalid_email					
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015		1 Other	user106@example.com					
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015		2	user176@example.com					
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015		1 Male	user168@example.com					
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015		1 Male	user49@example.com					
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015		2 Female	user24@example.com					
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015		Male	user168@example.com					
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015		2 Other	user26@example.com					
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015		Female	user25@example.com					
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015		Male	user153@example.com					
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015		1 Female	user18@example.com					
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015		5 Male	user1@example.com					
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015		5 Male	user106@example.com					
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015		4	user180@example.com					
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015		2 Male	user184@example.com					
EMP1015		IT	India	₹ 92,443.00	19-04-2015		3 Other	user3@example.com					
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015			user64@example.com					
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015		1 Other	user38@example.com					
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015			user96@example.com					
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015		5 Female	user151@example.com					
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015		2 Other	user200@example.com					
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015		4	user77@example.com					
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015		5	user128@example.com					
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015		3 Male	user149@example.com					
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015		1 Other	user122@example.com					
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015		3	user41@example.com					

Metric	Salary
SUM	₹ 94,11,892.00
AVERAGE	₹ 66,280.93
MIN	₹ 30,280.00
MAX	₹ 99,621.00
COUNT	142

Metric	PERFORMANCE RATING
SUM	359
AVERAGE	2.872
MIN	1
MAX	5
COUNT	125
COUNTBLANK	25

File Home Insert Draw Page Layout Formulas Data Review View Help										Comments		Share	
Cut Copy Format Painter Clipboard Font Alignment Number Styles Wrap Text Merge & Center Conditional Formatting Format as Table Cell Styles Insert Delete Format										AutoSum Fill Clear		Sort & Filter Find & Select Add-ins	
L17													
=MIN(G:G)													
Employee ID	Name	Department	Country	Salary	Date of Join	Performance Rating	Gender	Email					
EMP1000		Finance		N/A			2 Unknown	invalid_email					
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015		1 Other	user106@example.com					
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015		2	user176@example.com					
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015		1 Male	user168@example.com					
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015		1 Male	user49@example.com					
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015		2 Female	user24@example.com					
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015		Male	user168@example.com					
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015		2 Other	user26@example.com					
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015		Female	user25@example.com					
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015		Male	user153@example.com					
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015		1 Female	user18@example.com					
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015		5 Male	user1@example.com					
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015		5 Male	user106@example.com					
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015		4	user180@example.com					
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015		2 Male	user184@example.com					
EMP1015		IT	India	₹ 92,443.00	19-04-2015		3 Other	user3@example.com					
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015			user64@example.com					
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015		1 Other	user38@example.com					
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015			user96@example.com					
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015		5 Female	user151@example.com					
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015		2 Other	user200@example.com					
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015		4	user77@example.com					
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015		5	user128@example.com					
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015		3 Male	user149@example.com					
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015		1 Other	user122@example.com					
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015		3	user41@example.com					

Metric	Salary
SUM	₹ 94,11,892.00
AVERAGE	₹ 66,280.93
MIN	₹ 30,280.00
MAX	₹ 99,621.00
COUNT	142

Metric	PERFORMANCE RATING
SUM	359
AVERAGE	2.872
MIN	1
MAX	5
COUNT	125
COUNTBLANK	25

File

Home

Insert

Draw

Page Layout

Formulas

Data

Review

View

Help

Cut

Copy

Format Painter

Clipboard

Font

Alignment

Merge & Center

Number

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

AutoSum

Fill

Clear

Editing

Sort & Filter

Find & Select

Add-ins

Comments

Share

120

Day 4 – Basic Excel Formulas (Part 2)

Conditional functions such as **IF**, **COUNTIF**, **SUMIF**, and **AVERAGEIF** were used to analyze data based on specific conditions. These functions help in filtering and evaluating data efficiently, such as identifying high-performing employees or counting employees with salaries above a certain threshold.

Home

Insert

Draw

Page Layout

Formulas

Data

Review

View

Help

Table Design

Cut

Copy

Format Painter

Clipboard

Calibri

11

A⁺

A⁻

</

The **IF** function is used to check a condition and return different results based on whether the condition is true or false, such as identifying whether an employee is a high performer or not based on their performance rating.

Employee ID Name Department Country Salary Date of Joining Rating Gender Email LEFT RIGHT MID												
EMP1000		Finance		N/A		2	Unknown	invalid_email	EMP	1000	P100	
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	1	Other	user106@example.com	EMP	1001	P100	
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2		user176@example.com	EMP	1002	P100	
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com	EMP	1003	P100	
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com	EMP	1004	P100	
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com	EMP	1005	P100	
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015		Male	user168@example.com	EMP	1006	P100	
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015	2	Other	user26@example.com	EMP	1007	P100	
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015		Female	user25@example.com	EMP	1008	P100	
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015		Male	user153@example.com	EMP	1009	P100	
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com	EMP	1010	P101	
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com	EMP	1011	P101	
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015	5	Male	user106@example.com	EMP	1012	P101	
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015	4		user180@example.com	EMP	1013	P101	
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015	2	Male	user184@example.com	EMP	1014	P101	
EMP1015		IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com	EMP	1015	P101	
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015			user64@example.com	EMP	1016	P101	
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com	EMP	1017	P101	
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015			user96@example.com	EMP	1018	P101	
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com	EMP	1019	P101	
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015	2	Other	user200@example.com	EMP	1020	P102	
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015	4		user77@example.com	EMP	1021	P102	
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5		user128@example.com	EMP	1022	P102	
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com	EMP	1023	P102	
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015	1	Other	user122@example.com	EMP	1024	P102	
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015	3		user41@example.com	EMP	1025	P102	

MID

Employee ID Name Department Country Salary Date of Joining Rating Gender Email LEFT RIGHT MID												
EMP1000		Finance		N/A		2	Unknown	invalid_email	EMP	1000	P100	
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	1	Other	user106@example.com	EMP	1001	P100	
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2		user176@example.com	EMP	1002	P100	
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com	EMP	1003	P100	
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com	EMP	1004	P100	
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com	EMP	1005	P100	
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015		Male	user168@example.com	EMP	1006	P100	
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015	2	Other	user26@example.com	EMP	1007	P100	
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015		Female	user25@example.com	EMP	1008	P100	
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015		Male	user153@example.com	EMP	1009	P100	
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com	EMP	1010	P101	
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com	EMP	1011	P101	
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015	5	Male	user106@example.com	EMP	1012	P101	
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015	4		user180@example.com	EMP	1013	P101	
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015	2	Male	user184@example.com	EMP	1014	P101	
EMP1015		IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com	EMP	1015	P101	
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015			user64@example.com	EMP	1016	P101	
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com	EMP	1017	P101	
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015			user96@example.com	EMP	1018	P101	
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com	EMP	1019	P101	
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015	2	Other	user200@example.com	EMP	1020	P102	
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015	4		user77@example.com	EMP	1021	P102	
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5		user128@example.com	EMP	1022	P102	
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com	EMP	1023	P102	
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015	1	Other	user122@example.com	EMP	1024	P102	
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015	3		user41@example.com	EMP	1025	P102	

UPPER

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query																	Comments	Share
Clipboard Font Alignment Number Styles Cells Editing Add-ins																		
M2 =UPPER([@Country])																		
Employee ID	Name	Department	Country	Salary	Date of Joining	Ratio	Gender	Email	LEFT	RIGHT	MID	UPPER	LOWER	Prope				
EMP1000		Finance		N/A		2	Unknown	invalid_email	EMP	1000	P100							
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	1	Other	user106@example.com	EMP	1001	P100	GERMANY	germany	Germany				
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2		user176@example.com	EMP	1002	P100	GERMANY	germany	Germany				
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com	EMP	1003	P100	USA	usa	Usa				
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com	EMP	1004	P100	GERMANY	germany	Germany				
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com	EMP	1005	P100	INDIA	india	India				
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015		Male	user168@example.com	EMP	1006	P100							
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015	2	Other	user26@example.com	EMP	1007	P100							
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015		Female	user25@example.com	EMP	1008	P100	GERMANY	germany	Germany				
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015		Male	user153@example.com	EMP	1009	P100	UK	uk	Uk				
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com	EMP	1010	P101	UK	uk	Uk				
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com	EMP	1011	P101	USA	usa	Usa				
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015	5	Male	user106@example.com	EMP	1012	P101							
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015	4		user180@example.com	EMP	1013	P101	INDIA	india	India				
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015	2	Male	user184@example.com	EMP	1014	P101							
EMP1015		IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com	EMP	1015	P101	INDIA	india	India				
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015			user64@example.com	EMP	1016	P101	GERMANY	germany	Germany				
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com	EMP	1017	P101	AUSTRALIA	australia	Australia				
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015			user96@example.com	EMP	1018	P101	UK	uk	Uk				
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com	EMP	1019	P101	INDIA	india	India				
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015	2	Other	user200@example.com	EMP	1020	P102	USA	usa	Usa				
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015	4		user77@example.com	EMP	1021	P102							
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5		user128@example.com	EMP	1022	P102	INDIA	india	India				
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com	EMP	1023	P102	UK	uk	Uk				
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015	1	Other	user122@example.com	EMP	1024	P102							
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015	3		user41@example.com	EMP	1025	P102							

LOWER

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query																	Comments	Share
Clipboard Font Alignment Number Styles Cells Editing Add-ins																		
M2 =LOWER([@Country])																		
Employee ID	Name	Department	Country	Salary	Date of Joining	Ratio	Gender	Email	LEFT	RIGHT	MID	UPPER	LOWER	Prope				
EMP1000		Finance		N/A		2	Unknown	invalid_email	EMP	1000	P100							
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	1	Other	user106@example.com	EMP	1001	P100	GERMANY	germany	Germany				
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2		user176@example.com	EMP	1002	P100	GERMANY	germany	Germany				
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com	EMP	1003	P100	USA	usa	Usa				
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com	EMP	1004	P100	GERMANY	germany	Germany				
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com	EMP	1005	P100	INDIA	india	India				
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015		Male	user168@example.com	EMP	1006	P100							
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015	2	Other	user26@example.com	EMP	1007	P100							
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015		Female	user25@example.com	EMP	1008	P100	GERMANY	germany	Germany				
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015		Male	user153@example.com	EMP	1009	P100	UK	uk	Uk				
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com	EMP	1010	P101	UK	uk	Uk				
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com	EMP	1011	P101	USA	usa	Usa				
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015	5	Male	user106@example.com	EMP	1012	P101							
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015	4		user180@example.com	EMP	1013	P101	INDIA	india	India				
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015	2	Male	user184@example.com	EMP	1014	P101							
EMP1015		IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com	EMP	1015	P101	INDIA	india	India				
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015			user64@example.com	EMP	1016	P101	GERMANY	germany	Germany				
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com	EMP	1017	P101	AUSTRALIA	australia	Australia				
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015			user96@example.com	EMP	1018	P101	UK	uk	Uk				
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com	EMP	1019	P101	INDIA	india	India				
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015	2	Other	user200@example.com	EMP	1020	P102	USA	usa	Usa				
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015	4		user77@example.com	EMP	1021	P102							
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5		user128@example.com	EMP	1022	P102	INDIA	india	India				
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com	EMP	1023	P102	UK	uk	Uk				
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015	1	Other	user122@example.com	EMP	1024	P102							
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015	3		user41@example.com	EMP	1025	P102							

PROPER

The screenshot shows an Excel spreadsheet with a table of employee data. The formula bar at the top displays the formula `=PROPER([@LOWER])`, which is used to convert the text in the selected cell to proper case. The table includes columns for Employee ID, Name, Department, Country, Salary, Date of Joining, Rating, Gender, Email, and various performance metrics (LEFT, RIGHT, MID, UPPER, LOWER, Proper). The data is organized into rows, with each row representing an employee's record.

Employee ID	Name	Department	Country	Salary	Date of Joining	Rating	Gender	Email	LEFT	RIGHT	MID	UPPER	LOWER	Proper
EMP1000	Eve Adams	Finance	Germany	₹ 54,255.00	11-01-2015	2	Unknown	invalid_email	EMP	1000	P100	GERMANY	germany	Germany
EMP1001	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2	Other	user106@example.com	EMP	1001	P100	GERMANY	germany	Germany
EMP1002	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com	EMP	1002	P100	USA	usa	Usa
EMP1003	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com	EMP	1003	P100	GERMANY	germany	Germany
EMP1004	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com	EMP	1004	P100	INDIA	india	India
EMP1005	Sam Wilson	Marketing	USA	₹ 35,364.00	22-02-2015	2	Other	user26@example.com	EMP	1005	P100	USA	usa	Usa
EMP1006	Chris Lee	Marketing	Germany	₹ 97,881.00	01-03-2015	2	Female	user25@example.com	EMP	1006	P100	GERMANY	germany	Germany
EMP1007	Alice Brown	IT	UK	₹ 91,846.00	08-03-2015	1	Male	user153@example.com	EMP	1007	P100	UK	uk	Uk
EMP1008	Chris Lee	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com	EMP	1008	P100	UK	uk	Uk
EMP1009	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com	EMP	1009	P101	USA	usa	Usa
EMP1010	Bob Johnson	Sales	India	₹ 96,418.00	29-03-2015	5	Male	user106@example.com	EMP	1010	P101	INDIA	india	India
EMP1011	Sam Wilson	Finance	India	₹ 62,158.00	05-04-2015	4	Other	user180@example.com	EMP	1011	P101	INDIA	india	India
EMP1012	Jane Smith	HR	India	₹ 59,894.00	12-04-2015	2	Male	user184@example.com	EMP	1012	P101	INDIA	india	India
EMP1013	Jane Smith	IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com	EMP	1013	P101	INDIA	india	India
EMP1014	Bob Johnson	Finance	Germany	₹ 90,271.00	26-04-2015	1	Other	user64@example.com	EMP	1014	P101	GERMANY	germany	Germany
EMP1015	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com	EMP	1015	P101	AUSTRALIA	australia	Australia
EMP1016	Sam Wilson	Finance	UK	₹ 73,602.00	10-05-2015	2	Other	user96@example.com	EMP	1016	P101	UK	uk	Uk
EMP1017	Eve Adams	Finance	India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com	EMP	1017	P101	INDIA	india	India
EMP1018	Jane Smith	HR	USA	N/A	24-05-2015	2	Other	user200@example.com	EMP	1018	P102	USA	usa	Usa
EMP1019	Jane Smith	HR	India	₹ 98,922.00	31-05-2015	4	Other	user77@example.com	EMP	1019	P102	INDIA	india	India
EMP1020	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5	Male	user128@example.com	EMP	1020	P102	INDIA	india	India
EMP1021	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com	EMP	1021	P102	UK	uk	Uk
EMP1022	Sam Wilson	Finance	India	₹ 91,005.00	21-06-2015	1	Other	user122@example.com	EMP	1022	P102	INDIA	india	India
EMP1023	Chris Lee	Finance	India	₹ 52,272.00	28-06-2015	3	Other	user41@example.com	EMP	1023	P102	INDIA	india	India

Day 6 – Data Validation & Cleaning

Remove nulls, duplicates, replace inconsistent data

The screenshot shows the same Excel spreadsheet as before, but with the 'Data Validation' task pane open on the right side. The task pane is set to the 'Data Validation' tab, and the 'Allow' dropdown is set to 'List'. The 'Source' field is empty. The 'Data Validation' task pane is used to define rules for the data in the spreadsheet, such as restricting the values in a cell to a specific list or range.

Employee ID	Name	Department	Country	Salary	Date of Joining	Rating	Gender	Email
EMP1000	Eve Adams	Finance	Germany	₹ 54,255.00	11-01-2015	2	Unknown	invalid_email
EMP1001	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2	Other	user106@example.com
EMP1002	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com
EMP1003	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com
EMP1004	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com
EMP1005	Sam Wilson	Marketing	USA	₹ 35,364.00	22-02-2015	2	Other	user26@example.com
EMP1006	Chris Lee	Marketing	Germany	₹ 97,881.00	01-03-2015	2	Female	user25@example.com
EMP1007	Alice Brown	IT	UK	₹ 91,846.00	08-03-2015	1	Male	user153@example.com
EMP1008	Chris Lee	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com
EMP1009	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com
EMP1010	Bob Johnson	Sales	India	₹ 96,418.00	29-03-2015	5	Male	user106@example.com
EMP1011	Sam Wilson	Finance	India	₹ 62,158.00	05-04-2015	4	Other	user180@example.com
EMP1012	Jane Smith	HR	India	₹ 59,894.00	12-04-2015	2	Male	user184@example.com
EMP1013	Jane Smith	IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com
EMP1014	Bob Johnson	Finance	Germany	₹ 90,271.00	26-04-2015	1	Other	user64@example.com
EMP1015	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com
EMP1016	Sam Wilson	Finance	UK	₹ 73,602.00	10-05-2015	2	Other	user96@example.com
EMP1017	Eve Adams	Finance	India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com
EMP1018	Jane Smith	HR	USA	N/A	24-05-2015	2	Other	user200@example.com
EMP1019	Jane Smith	HR	India	₹ 98,922.00	31-05-2015	4	Other	user77@example.com
EMP1020	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5	Male	user128@example.com
EMP1021	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com
EMP1022	Sam Wilson	Finance	India	₹ 91,005.00	21-06-2015	1	Other	user122@example.com
EMP1023	Chris Lee	Finance	India	₹ 52,272.00	28-06-2015	3	Other	user41@example.com

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query

Clipboard Font Alignment Number Styles Cells Editing Add-ins

Employee ID

Employee ID	Name	Department	Country	Salary	Date of Joining	Rating	Gender	Email
EMP1000		Finance		N/A		2	Unknown	invalid_email
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2		
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015		Male	user168@example.com
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015	2	Other	user26@example.com
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015		Female	user25@example.com
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015		Male	user153@example.com
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015	5	Male	user106@example.com
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015	4		user180@example.com
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015	2	Male	user184@example.com
EMP1015		IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015			user64@example.com
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015			user96@example.com
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015	2	Other	user200@example.com
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015	4		user77@example.com
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5		user128@example.com
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015	1	Other	user122@example.com
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015	3		user41@example.com

DAY3-SUM,AVERAGE,COUNT DAY4-IF conditions DAY5-TEXT Functions DAY6-DATACLEANING

Duplicate values were checked using conditional formatting . No duplicate records were found in the Employee ID column, ensuring that each record is unique.

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query

Clipboard Font Alignment Number Styles Cells Editing Add-ins

Email

Employee ID	Name	Department	Country	Salary	Date of Joining	Rating	Gender	Email
EMP1000		Finance		N/A		2	Unknown	invalid_email
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2		
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-02-2015		Male	user168@example.com
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-02-2015	2	Other	user26@example.com
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015		Female	user25@example.com
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015		Male	user153@example.com
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-03-2015	5	Male	user106@example.com
EMP1013	Sam Wilson		India	₹ 62,158.00	05-04-2015	4		user180@example.com
EMP1014	Jane Smith	HR		₹ 59,894.00	12-04-2015	2	Male	user184@example.com
EMP1015		IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015			user64@example.com
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015			user96@example.com
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015	2	Other	user200@example.com
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015	4		user77@example.com
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5		user128@example.com
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015	1	Other	user122@example.com
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015	3		user41@example.com

DAY6-DATACLEANING

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design

Get Data From Text/CSV Recent Sources Queries & Connections Refresh All Workbook Links

Sort & Filter Filter Reapply Advanced Text to Columns Remove Duplicates Data Validation Data Tools Consolidate Data Model What-If Analysis Forecast Sheet Group Ungroup Subtotal Outline Analysis

Remove Duplicates

To delete duplicate values, select one or more columns that contain duplicates.

Select All Unselect All My data has headers

Columns

Salary

Date of Joining

Performance Rating

Gender

Email

OK Cancel

Employee ID	Name	Department	Country	Salary	Date of Jo
EMP1000		Finance		N/A	
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-03-2015
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-03-2015
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-04-2015
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-04-2015
EMP1006	Sam Wilson	Sales		₹ 67,721.00	15-04-2015
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-04-2015
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-05-2015
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-05-2015
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-05-2015
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-05-2015
EMP1012	Bob Johnson	Sales		₹ 96,418.00	29-05-2015
EMP1013	Sam Wilson		India	₹ 62,158.00	05-06-2015
EMP1014	Jane Smith	HR		₹ 59,894.00	12-06-2015
EMP1015	Jane Smith	IT	India	₹ 92,443.00	19-04-2015
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015

DAY3-SUM,AVERAGE,COUNT DAY4-IF conditions DAY5-TEXT Functions DAY6-DATACLEANING

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design

Get Data From Text/CSV Recent Sources Queries & Connections Refresh All Workbook Links

Sort & Filter Filter Reapply Advanced Text to Columns Remove Duplicates Data Validation Data Tools Consolidate Data Model What-If Analysis Forecast Sheet Group Ungroup Subtotal Outline Analysis

Microsoft Excel

51 duplicate values found and removed; 99 unique values remain. Note that counts may include empty cells, spaces, etc.

OK

Employee ID	Name	Department	Country	Salary	Date of Joining	Rating	Gender	Email
EMP1000		Finance		N/A				
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	3	Unknown	invalid_email
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-03-2015	3	Other	user106@example.com
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-03-2015	3	Other	user176@example.com
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-04-2015	3	Other	user176@example.com
EMP1005	Alice Brown	IT	India	₹ 78,421.00	08-04-2015	3	Other	user176@example.com
EMP1007	Chris Lee	Marketing		₹ 35,364.00	22-04-2015	3	Other	user176@example.com
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00	01-05-2015	3	Other	user176@example.com
EMP1009	Chris Lee	HR	UK	₹ 91,846.00	08-05-2015	3	Other	user176@example.com
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00	15-05-2015	3	Other	user176@example.com
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00	22-05-2015	3	Other	user176@example.com
EMP1013	Sam Wilson		India	₹ 62,158.00	05-06-2015	3	Other	user176@example.com
EMP1014	Jane Smith	HR		₹ 59,894.00	12-06-2015	3	Other	user176@example.com
EMP1015	Jane Smith	IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-04-2015	3	Other	user64@example.com
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015	3	Other	user96@example.com
EMP1019	Eve Adams		India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com
EMP1020	Jane Smith	HR	USA	N/A	24-05-2015	2	Other	user200@example.com
EMP1021	Jane Smith	HR		₹ 98,922.00	31-05-2015	4	Other	user77@example.com
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5	Other	user128@example.com
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015	1	Other	user122@example.com
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015	3	Other	user41@example.com
EMP1026	Eve Adams	Sales	Germany	₹ 76,518.00	05-07-2015	2	Female	user68@example.com
EMP1027	Alice Brown	IT	UK	₹ 41,448.00	12-07-2015	3	Other	user103@example.com

DAY3-SUM,AVERAGE,COUNT DAY4-IF conditions DAY5-TEXT Functions DAY6-DATACLEANING

Duplicate email values were identified and handled by using filtering and conditional formatting techniques.

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query

Get From Text/CSV Recent Sources From Web Existing Connections Refresh All Workbook Links Queries & Connections Sort & Filter Filter Clear Reapply Advanced Text to Columns Flash Fill Remove Duplicates Consolidate Data Model What-If Analysis Forecast Sheet Group Ungroup Subtotal Outline Analysis

Salary

Employee ID	Name	Department	Country	Salary	Date of Joining	Ratio	Gender	Email
EMP1000		Finance				2	Unknown	invalid_email
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2		user176@example.com
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com
EMP1005	Alice Brown	IT	India	₹ 78,421.00				
EMP1007	Chris Lee	Marketing		₹ 35,364.00				
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00				
EMP1009	Chris Lee	HR	UK	₹ 91,846.00				
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00				
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00				
EMP1013	Sam Wilson		India	₹ 62,158.00				
EMP1014	Jane Smith	HR		₹ 59,894.00				
EMP1015		IT	India	₹ 92,443.00				
EMP1016	Bob Johnson		Germany	₹ 90,271.00	26-06-2015			
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-07-2015			
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-07-2015			
EMP1019	Eve Adams		India	₹ 67,531.00	17-07-2015			
EMP1020	Jane Smith	HR	USA	₹ 98,922.00	31-07-2015			
EMP1021	Jane Smith	HR		₹ 98,922.00	31-07-2015			
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5		user128@example.com
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015	1	Other	user122@example.com
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015	3		user41@example.com
EMP1026	Eve Adams	Sales	Germany	₹ 76,518.00	05-07-2015	2	Female	user68@example.com
EMP1027	Alice Brown	IT	UK	₹ 41,448.00	12-07-2015	3	Other	user103@example.com

Find and Replace

Find what: N/A

Replace with:

Replace all

Microsoft Excel

All done. We made 6 replacements.

OK

DAY3-SUM.AVERAGE.COUNT DAY4-IF conditions DAY5-TEXT Functions DAY6-DATACLEANING

Ready Accessibility: Investigate

Average: ₹ 68,269.25 Count: 100 Sum: ₹ 63,49,040.00

The missing name are filled by unknown

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query

Get From Text/CSV Recent Sources From Web Existing Connections Refresh All Workbook Links Queries & Connections Sort & Filter Filter Clear Reapply Advanced Text to Columns Flash Fill Remove Duplicates Consolidate Data Model What-If Analysis Forecast Sheet Group Ungroup Subtotal Outline Analysis

Name

Employee ID	Name	Department	Country	Salary	Date of Joining	Ratio	Gender	Email
EMP1000	Not Available	Finance				2	Unknown	invalid_email
EMP1001	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2		user176@example.com
EMP1003	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com
EMP1004	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com
EMP1005	Alice Brown	IT	India	₹ 78,421.00				
EMP1007	Chris Lee	Marketing		₹ 35,364.00				
EMP1008	Alice Brown	IT	Germany	₹ 97,881.00				
EMP1009	Chris Lee	HR	UK	₹ 91,846.00				
EMP1010	Sam Wilson	Finance	UK	₹ 64,247.00				
EMP1011	Sam Wilson	IT	USA	₹ 91,180.00				
EMP1013	Sam Wilson		India	₹ 62,158.00				
EMP1014	Jane Smith	HR		₹ 59,894.00				
EMP1015	Not Available	IT	India	₹ 92,443.00				
EMP1016	Bob Johnson		Germany	₹ 90,271.00				
EMP1017	Jane Smith	Finance	Australia	₹ 41,341.00	03-07-2015			
EMP1018	Alice Brown	Finance	UK	₹ 73,602.00	10-07-2015			
EMP1019	Eve Adams		India	₹ 67,531.00	17-07-2015			
EMP1020	Jane Smith	HR	USA	₹ 98,922.00	31-07-2015			
EMP1021	Jane Smith	HR		₹ 98,922.00	31-07-2015			
EMP1022	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5		user128@example.com
EMP1023	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Finance		₹ 91,005.00	21-06-2015	1	Other	user122@example.com
EMP1025	Chris Lee			₹ 52,272.00	28-06-2015	3		user41@example.com
EMP1026	Eve Adams	Sales	Germany	₹ 76,518.00	05-07-2015	2	Female	user68@example.com
EMP1027	Alice Brown	IT	UK	₹ 41,448.00	12-07-2015	3	Other	user103@example.com

Find and Replace

Find what: Not Available

Replace with:

Replace all

Microsoft Excel

All done. We made 57 replacements.

OK

DAY3-SUM.AVERAGE.COUNT DAY4-IF conditions DAY5-TEXT Functions DAY6-DATACLEANING

Ready Accessibility: Investigate

Count: 94

Employee ID	Name	PROPER	Department	Country	Salary	Date of Join	Ratio	Gender	Email
EMP1001	Eve Adams	Eve Adams	HR	Germany	₹ 54,255.00	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Alice Brown	Finance	Germany	₹ 63,184.00	18-01-2015	2	Not specif	user176@example.com
EMP1003	John Doe	John Doe	Sales	USA	₹ 94,406.00	25-01-2015	1	Male	user168@example.com
EMP1004	Jane Smith	Jane Smith	Sales	Germany	₹ 64,120.00	01-02-2015	1	Male	user49@example.com
EMP1005	Alice Brown	Alice Brown	IT	India	₹ 78,421.00	08-02-2015	2	Female	user24@example.com
EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	₹ 35,364.00	22-02-2015	2	Other	user26@example.com
EMP1008	Alice Brown	Alice Brown	IT	Germany	₹ 97,881.00	01-03-2015	Unknown	Female	user25@example.com
EMP1009	Chris Lee	Chris Lee	HR	UK	₹ 91,846.00	08-03-2015	Unknown	Male	user153@example.com
EMP1010	Sam Wilson	Sam Wilson	Finance	UK	₹ 64,247.00	15-03-2015	1	Female	user18@example.com
EMP1011	Sam Wilson	Sam Wilson	IT	USA	₹ 91,180.00	22-03-2015	5	Male	user1@example.com
EMP1013	Sam Wilson	Sam Wilson	Unknown	India	₹ 62,158.00	05-04-2015	4	Not specif	user180@example.com
EMP1014	Jane Smith	Jane Smith	HR	Unknown	₹ 59,894.00	12-04-2015	2	Male	user184@example.com
EMP1015	Unknown	Unknown	IT	India	₹ 92,443.00	19-04-2015	3	Other	user3@example.com
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	₹ 90,271.00	26-04-2015	Unknown	Not specif	user64@example.com
EMP1017	Jane Smith	Jane Smith	Finance	Australia	₹ 41,341.00	03-05-2015	1	Other	user38@example.com
EMP1018	Alice Brown	Alice Brown	Finance	UK	₹ 73,602.00	10-05-2015	Unknown	Not specif	user96@example.com
EMP1019	Eve Adams	Eve Adams	Unknown	India	₹ 67,531.00	17-05-2015	5	Female	user151@example.com
EMP1020	Jane Smith	Jane Smith	HR	USA	₹ 68,269.25	24-05-2015	2	Other	user200@example.com
EMP1021	Jane Smith	Jane Smith	HR	Unknown	₹ 98,922.00	31-05-2015	4	Not specif	user77@example.com
EMP1022	Eve Adams	Eve Adams	Finance	India	₹ 38,333.00	07-06-2015	5	Not specif	user128@example.com
EMP1023	Alice Brown	Alice Brown	Marketing	UK	₹ 36,515.00	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	₹ 91,005.00	21-06-2015	1	Other	user122@example.com
EMP1025	Chris Lee	Chris Lee	Unknown	Unknown	₹ 52,722.00	28-06-2015	3	Not specif	user41@example.com
EMP1026	Eve Adams	Eve Adams	Sales	Germany	₹ 76,518.00	05-07-2015	2	Female	user68@example.com
EMP1027	Alice Brown	Alice Brown	IT	UK	₹ 41,448.00	12-07-2015	3	Other	user103@example.com
EMP1028	John Doe	John Doe	HR	UK	₹ 49,282.00	19-07-2015	2	Female	user135@example.com

The final dataset after data cleaning the salary column missing values are filled by calculating the average salary.

Day 7 – Power Query Introduction:

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Join
EMP1001	Eve Adams	Eve Adams	HR	Germany	54255	11
EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18
EMP1003	John Doe	John Doe	Sales	USA	94406	25
EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01
EMP1005	Alice Brown	Alice Brown	IT	India	78421	08
EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	35364	22
EMP1008	Alice Brown	Alice Brown	IT	Germany	97881	01
EMP1009	Chris Lee	Chris Lee	HR	UK	91846	08
EMP1010	Sam Wilson	Sam Wilson	Finance	UK	64247	15
EMP1011	Sam Wilson	Sam Wilson	IT	USA	91180	22
EMP1013	Sam Wilson	Sam Wilson	Unknown	India	62158	05
EMP1014	Jane Smith	Jane Smith	HR	Unknown	59894	12
EMP1015	Unknown	Unknown	IT	India	92443	19
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26
EMP1017	Jane Smith	Jane Smith	Finance	Australia	41341	03
EMP1018	Alice Brown	Alice Brown	Finance	UK	73602	10
EMP1019	Eve Adams	Eve Adams	Unknown	India	67531	17
EMP1020	Jane Smith	Jane Smith	HR	USA	68269.25	24
EMP1021	Jane Smith	Jane Smith	HR	Unknown	98922	31
EMP1022	Eve Adams	Eve Adams	Finance	India	38333	07
EMP1023	Alice Brown	Alice Brown	Marketing	UK	36515	14
EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21
EMP1025	Chris Lee	Chris Lee	Unknown	Unknown	52722	28
EMP1026	Eve Adams	Eve Adams	Sales	Germany	76518	05
EMP1027	Alice Brown	Alice Brown	IT	UK	41448	12
EMP1028	John Doe	John Doe	HR	UK	49282	19
EMP1028	Bob Johnson	Bob Johnson	Marketing	UK	92959	26

Loaded data into power query.

File Home Transform Add Column View

Close & Load Refresh Preview Manage Query

Advanced Editor

Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By

Data Type: Currency * Use First Row as Headers * Replace Values

Merge Queries * Append Queries * Combine Files * Manage Parameters * Data source settings * New Source * Recent Sources * Enter Data * New Query

Table.TransformColumnTypes(Source,({"Employee ID", type text}, {"Name", type text}, {"PROPER NAMES", type text}, {"Department", type text}, {"Country", type text}, {"Salary", type number}, {"Date of Join", type date}))

EMPLOYEE ID	NAME	PROPER NAMES	Department	Country	Salary	Date of Join
1	EMP1001	Eve Adams	HR	Germany	54,255.00	11
2	EMP1002	Alice Brown	Finance	Germany	63,184.00	18
3	EMP1003	John Doe	Sales	USA	94,406.00	25
4	EMP1004	Jane Smith	Sales	Germany	64,120.00	01
5	EMP1005	Alice Brown	IT	India	78,421.00	08
6	EMP1007	Chris Lee	Marketing	Unknown	35,364.00	22
7	EMP1008	Alice Brown	IT	Germany	97,881.00	01
8	EMP1009	Chris Lee	HR	UK	91,846.00	08
9	EMP1010	Sam Wilson	Finance	UK	64,247.00	15
10	EMP1011	Sam Wilson	IT	USA	91,180.00	22
11	EMP1013	Sam Wilson	Unknown	India	62,158.00	05
12	EMP1014	Jane Smith	HR	Unknown	59,894.00	12
13	EMP1015	Unknown	IT	India	92,443.00	19
14	EMP1016	Bob Johnson	Unknown	Germany	90,271.00	26
15	EMP1017	Jane Smith	Finance	Australia	41,341.00	03
16	EMP1018	Alice Brown	Finance	UK	73,602.00	10
17	EMP1019	Eve Adams	Unknown	India	67,531.00	17
18	EMP1020	Jane Smith	HR	USA	68,269.25	24
19	EMP1021	Jane Smith	HR	Unknown	98,922.00	31
20	EMP1022	Eve Adams	Finance	India	38,333.00	07
21	EMP1023	Alice Brown	Marketing	UK	36,515.00	14
22	EMP1024	Sam Wilson	Finance	Unknown	91,005.00	21
23	EMP1025	Chris Lee	Unknown	Unknown	52,272.00	28
24	EMP1026	Eve Adams	Sales	Germany	76,518.00	05
25	EMP1027	Alice Brown	IT	UK	41,448.00	12
26	EMP1028	John Doe	HR	UK	49,282.00	19
27	EMP1029	Bob Johnson	Marketing	UK	92,959.00	26

10 COLUMNS, 98 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 06:45 PM

File Home Transform Add Column View

Group By Use First Row as Headers * Count Rows

Transpose Reverse Rows Reverse Data Type * Detect Data Type * Fill * Rename * Pivot Column * Move * Convert to List

Replace Values * Unpivot Columns * Split Column * Format * Extract * Parse * Lowercase * Uppercase * Capitalize Each Word

Merge Columns * Statistics Standard Scientific * Trigonometry * Rounding * Information * Date * Time * Duration * Expand * Aggregate * Extract Values * Create Data Type

Table.TransformColumnTypes(Source,({"Employee ID", type text}, {"Name", type text}, {"PROPER NAMES", type text}, {"Department", type text}, {"Country", type text}, {"Salary", type number}, {"Date of Join", type date}))

EMPLOYEE ID	NAME	PROPER NAMES	Department	Country	Salary	Date of Join
1	EMP1001	Eve Adams	HR	Germany	54,255.00	11
2	EMP1002	Alice Brown	Finance	Germany	63,184.00	18
3	EMP1003	John Doe	Sales	USA	94,406.00	25
4	EMP1004	Jane Smith	Sales	Germany	64,120.00	01
5	EMP1005	Alice Brown	IT	India	78,421.00	08
6	EMP1007	Chris Lee	Marketing	Unknown	35,364.00	22
7	EMP1008	Alice Brown	IT	Germany	97,881.00	01
8	EMP1009	Chris Lee	HR	UK	91,846.00	08
9	EMP1010	Sam Wilson	Finance	UK	64,247.00	15
10	EMP1011	Sam Wilson	IT	USA	91,180.00	22
11	EMP1013	Sam Wilson	Unknown	India	62,158.00	05
12	EMP1014	Jane Smith	HR	Unknown	59,894.00	12
13	EMP1015	Unknown	IT	India	92,443.00	19
14	EMP1016	Bob Johnson	Unknown	Germany	90,271.00	26
15	EMP1017	Jane Smith	Finance	Australia	41,341.00	03
16	EMP1018	Alice Brown	Finance	UK	73,602.00	10
17	EMP1019	Eve Adams	Unknown	India	67,531.00	17
18	EMP1020	Jane Smith	HR	USA	68,269.25	24
19	EMP1021	Jane Smith	HR	Unknown	98,922.00	31
20	EMP1022	Eve Adams	Finance	India	38,333.00	07
21	EMP1023	Alice Brown	Marketing	UK	36,515.00	14
22	EMP1024	Sam Wilson	Finance	Unknown	91,005.00	21
23	EMP1025	Chris Lee	Unknown	Unknown	52,272.00	28
24	EMP1026	Eve Adams	Sales	Germany	76,518.00	05
25	EMP1027	Alice Brown	IT	UK	41,448.00	12
26	EMP1028	John Doe	HR	UK	49,282.00	19
27	EMP1029	Bob Johnson	Marketing	UK	92,959.00	26

10 COLUMNS, 98 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 06:45 PM

Power Query Editor interface showing a query named "Trimmed Text3". The data table has 10 columns and 98 rows. The columns are: ID, Country, Salary, Date of Joining, Performance Rating, Gender, and Email. The data is displayed in a grid with alternating green and white rows. The right sidebar shows the "Query Settings" pane with "PROPERTIES" and "APPLIED STEPS" sections. The "APPLIED STEPS" list includes: Source, Changed Type, Trimmed Text, Trimmed Text1, Capitalized Each Word, Trimmed Text2, Capitalized Each Word1, Trimmed Text3, and Removed Duplicates.

ID	Country	Salary	Date of Joining	Performance Rating	Gender	Email
1	Germany	54,255.00	11-01-2015 00:00:00	1	Other	user106@example.com
2	Germany	63,184.00	18-01-2015 00:00:00	2	Not specified	user176@example.com
3	USA	94,406.00	25-01-2015 00:00:00	1	Male	user168@example.com
4	Germany	64,120.00	01-02-2015 00:00:00	1	Male	user49@example.com
5	India	78,421.00	08-02-2015 00:00:00	2	Female	user24@example.com
6	Unknown	35,364.00	22-02-2015 00:00:00	2	Other	user26@example.com
7	Germany	97,881.00	01-03-2015 00:00:00	Unknown	Female	user25@example.com
8	UK	91,846.00	08-03-2015 00:00:00	Unknown	Male	user153@example.com
9	UK	64,247.00	15-03-2015 00:00:00	1	Female	user18@example.com
10	USA	91,180.00	22-03-2015 00:00:00	5	Male	user1@example.com
11	India	62,158.00	05-04-2015 00:00:00	4	Not specified	user180@example.com
12	Unknown	59,894.00	12-04-2015 00:00:00	2	Male	user184@example.com
13	India	92,443.00	19-04-2015 00:00:00	3	Other	user3@example.com
14	Germany	90,271.00	26-04-2015 00:00:00	Unknown	Not specified	user64@example.com
15	Australia	41,341.00	03-05-2015 00:00:00	1	Other	user38@example.com
16	UK	73,602.00	10-05-2015 00:00:00	Unknown	Not specified	user96@example.com
17	India	67,531.00	17-05-2015 00:00:00	5	Female	user151@example.com
18	USA	68,269.25	24-05-2015 00:00:00	2	Other	user200@example.com
19	Unknown	98,922.00	31-05-2015 00:00:00	4	Not specified	user77@example.com
20	India	38,333.00	07-06-2015 00:00:00	5	Not specified	user128@example.com
21	UK	36,515.00	14-06-2015 00:00:00	3	Male	user149@example.com
22	Unknown	91,005.00	21-06-2015 00:00:00	1	Other	user122@example.com
23		52,272.00	28-06-2015 00:00:00	3	Not specified	user41@example.com
24	Germany	76,518.00	05-07-2015 00:00:00	2	Female	user68@example.com
25	UK	41,448.00	12-07-2015 00:00:00	3	Other	user103@example.com
26	UK	49,282.00	19-07-2015 00:00:00	2	Female	user135@example.com
27	UK	92,959.00	26-07-2015 00:00:00	5	Not specified	user83@example.com

Microsoft Excel interface showing the cleaned dataset loaded back into a worksheet. The data is displayed in a grid with columns A through Q. The columns are: Employee ID, Name, PROPER NAMES, Department, Country, Salary, Date of Joining, Performance Rating, Gender, and Email. The data is displayed in a grid with alternating green and white rows. The bottom status bar shows "Ready" and "Accessibility: Investigate".

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email
EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015	2	Not specified	user176@example.com
EMP1003	John Doe	John Doe	Sales	USA	94406	25-01-2015	1	Male	user168@example.com
EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015	1	Male	user49@example.com
EMP1005	Alice Brown	Alice Brown	It	India	78421	08-02-2015	2	Female	user24@example.com
EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	35364	22-02-2015	2	Other	user26@example.com
EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015	0	Female	user25@example.com
EMP1009	Chris Lee	Chris Lee	Hr	UK	91846	08-03-2015	0	Male	user153@example.com
EMP1010	Sam Wilson	Sam Wilson	Finance	UK	64247	15-03-2015	1	Female	user18@example.com
EMP1011	Sam Wilson	Sam Wilson	It	USA	91180	22-03-2015	5	Male	user1@example.com
EMP1013	Sam Wilson	Sam Wilson	Unknown	India	62158	05-04-2015	4	Not specified	user180@example.com
EMP1014	Jane Smith	Jane Smith	Hr	Unknown	59894	12-04-2015	2	Male	user184@example.com
EMP1015	Unknown	Unknown	It	India	92443	19-04-2015	3	Other	user3@example.com
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26-04-2015	0	Not specified	user64@example.com
EMP1017	Jane Smith	Jane Smith	Finance	Australia	41341	03-05-2015	1	Other	user38@example.com
EMP1018	Alice Brown	Alice Brown	Finance	UK	73602	10-05-2015	0	Not specified	user96@example.com
EMP1019	Eve Adams	Eve Adams	Unknown	India	67531	17-05-2015	5	Female	user151@example.com
EMP1020	Jane Smith	Jane Smith	Hr	USA	68269	24-05-2015	2	Other	user200@example.com
EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015	4	Not specified	user77@example.com
EMP1022	Eve Adams	Eve Adams	Finance	India	38333	07-06-2015	5	Not specified	user128@example.com
EMP1023	Alice Brown	Alice Brown	Marketing	UK	36515	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21-06-2015	1	Other	user122@example.com
EMP1025	Chris Lee	Chris Lee	Unknown	Unknown	52272	28-06-2015	3	Not specified	user41@example.com
EMP1026	Eve Adams	Eve Adams	Sales	Germany	76518	05-07-2015	2	Female	user68@example.com
EMP1027	Alice Brown	Alice Brown	It	UK	41448	12-07-2015	3	Other	user103@example.com
EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015	2	Female	user135@example.com

The dataset was loaded into Power Query using the “From Table/Range” option. Data cleaning operations such as replacing missing values, correcting invalid entries, trimming extra spaces, and standardizing text were performed within the Power Query Editor. Duplicate values were also handled. The cleaned dataset was then loaded back into Excel using the “Close & Load” option.

Day 8 – Handling Inconsistent Data

The screenshot shows the 'Convert Text to Columns Wizard - Step 1 of 3' dialog box in Microsoft Excel. The dialog is open over a spreadsheet. The 'Original data type' is 'Delimited'. The 'Text Wizard' has determined that the data is delimited. The 'Preview of selected data' shows a list of employee names. The background spreadsheet has columns A through Q, with data for employees and their details.

Convert Text to Columns Wizard - Step 1 of 3

The Text Wizard has determined that your data is Delimited.
If this is correct, choose Next, or choose the data type that best describes your data.

Original data type

Choose the file type that best describes your data:

- ☒ Delimited - Characters such as commas or tabs separate each field.
- ☐ Fixed width - Fields are aligned in columns with spaces between each field.

Preview of selected data:

Cancel < Back Next > Finish

Background Spreadsheet Data:

Employee ID	Name	Gender	Email
EMP1001	Eve Adams	1 Other	user106@example.com
EMP1002	Alice Brown	2 Not specified	user176@example.com
EMP1003	John Doe	1 Male	user168@example.com
EMP1004	Jane Smith	1 Male	user49@example.com
EMP1005	Alice Brown	2 Female	user24@example.com
EMP1007	Chris Lee	2 Other	user26@example.com
EMP1008	Alice Brown	0 Female	user25@example.com
EMP1009	Chris Lee	0 Male	user153@example.com
EMP1010	Sam Wilson	1 Female	user18@example.com
EMP1011	Sam Wilson	5 Male	user1@example.com
EMP1013	Sam Wilson	4 Not specified	user180@example.com
EMP1014	Jane Smith	2 Male	user184@example.com
EMP1015	Unknown	3 Other	user3@example.com
EMP1016	Bob Johnson	0 Not specified	user64@example.com
EMP1017	Jane Smith	1 Other	user38@example.com
EMP1018	Alice Brown	0 Not specified	user96@example.com
EMP1019	Eve Adams	5 Female	user151@example.com
EMP1020	Jane Smith	2 Other	user200@example.com
EMP1021	Jane Smith	4 Not specified	user77@example.com
EMP1022	Eve Adams	5 Not specified	user128@example.com
EMP1023	Alice Brown	3 Male	user149@example.com
EMP1024	Sam Wilson	1 Other	user122@example.com
EMP1025	Chris Lee	3 Not specified	user41@example.com
EMP1026	Eve Adams	2 Female	user68@example.com
EMP1027	Alice Brown	3 Other	user103@example.com
EMP1028	John Doe	2 Female	user135@example.com

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email	First Name	Last Name
EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015	1	Other	user106@example.com	Eve	Adams
EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015	2	Not specified	user176@example.com	Alice	Brown
EMP1003	John Doe	John Doe	Sales	USA	94406	25-01-2015	1	Male	user168@example.com	John	Doe
EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015	1	Male	user49@example.com	Jane	Smith
EMP1005	Alice Brown	Alice Brown	It	India	78421	08-02-2015	2	Female	user24@example.com	Alice	Brown
EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	35364	22-02-2015	2	Other	user26@example.com	Chris	Lee
EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015	0	Female	user25@example.com	Alice	Brown
EMP1009	Chris Lee	Chris Lee	Hr	UK	91846	08-03-2015	0	Male	user153@example.com	Chris	Lee
EMP1010	Sam Wilson	Sam Wilson	Finance	UK	64247	15-03-2015	1	Female	user18@example.com	Sam	Wilson
EMP1011	Sam Wilson	Sam Wilson	It	USA	91180	22-03-2015	5	Male	user1@example.com	Sam	Wilson
EMP1013	Sam Wilson	Sam Wilson	Unknown	India	62158	05-04-2015	4	Not specified	user180@example.com	Sam	Wilson
EMP1014	Jane Smith	Jane Smith	Hr	Unknown	59894	12-04-2015	2	Male	user184@example.com	Jane	Smith
EMP1015	Unknown	Unknown	It	India	92443	19-04-2015	3	Other	user3@example.com	Unknown	Unknown
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26-04-2015	0	Not specified	user64@example.com	Bob	Johnson
EMP1017	Jane Smith	Jane Smith	Finance	Australia	41341	03-05-2015	1	Other	user38@example.com	Jane	Smith
EMP1018	Alice Brown	Alice Brown	Finance	UK	73602	10-05-2015	0	Not specified	user96@example.com	Alice	Brown
EMP1019	Eve Adams	Eve Adams	Unknown	India	67531	17-05-2015	5	Female	user151@example.com	Eve	Adams
EMP1020	Jane Smith	Jane Smith	Hr	USA	68269	24-05-2015	2	Other	user200@example.com	Jane	Smith
EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015	4	Not specified	user77@example.com	Jane	Smith
EMP1022	Eve Adams	Eve Adams	Finance	India	38333	07-06-2015	5	Not specified	user128@example.com	Eve	Adams
EMP1023	Alice Brown	Alice Brown	Marketing	UK	36515	14-06-2015	3	Male	user149@example.com	Alice	Brown
EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21-06-2015	1	Other	user122@example.com	Sam	Wilson
EMP1025	Chris Lee	Chris Lee	Unknown	Unknown	52272	28-06-2015	3	Not specified	user41@example.com	Chris	Lee
EMP1026	Eve Adams	Eve Adams	Sales	Germany	76518	05-07-2015	2	Female	user68@example.com	Eve	Adams
EMP1027	Alice Brown	Alice Brown	It	UK	41448	12-07-2015	3	Other	user103@example.com	Alice	Brown
EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015	2	Female	user135@example.com	John	Doe

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email	First Name	Last Name	FF
EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015	1	Other	user106@example.com	Eve	Adams	Eve
EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015	2	Not specified	user176@example.com	Alice	Brown	Alice
EMP1003	John Doe	John Doe	Sales	USA	94406	25-01-2015	1	Male	user168@example.com	John	Doe	John
EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015	1	Male	user49@example.com	Jane	Smith	Jane
EMP1005	Alice Brown	Alice Brown	It	India	78421	08-02-2015	2	Female	user24@example.com	Alice	Brown	Alice
EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	35364	22-02-2015	2	Other	user26@example.com	Chris	Lee	Chris
EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015	0	Female	user25@example.com	Alice	Brown	Alice
EMP1009	Chris Lee	Chris Lee	Hr	UK	91846	08-03-2015	0	Male	user153@example.com	Chris	Lee	Chris
EMP1010	Sam Wilson	Sam Wilson	Finance	UK	64247	15-03-2015	1	Female	user18@example.com	Sam	Wilson	Sam
EMP1011	Sam Wilson	Sam Wilson	It	USA	91180	22-03-2015	5	Male	user1@example.com	Sam	Wilson	Sam
EMP1013	Sam Wilson	Sam Wilson	Unknown	India	62158	05-04-2015	4	Not specified	user180@example.com	Sam	Wilson	Sam
EMP1014	Jane Smith	Jane Smith	Hr	Unknown	59894	12-04-2015	2	Male	user184@example.com	Jane	Smith	Jane
EMP1015	Unknown	Unknown	It	India	92443	19-04-2015	3	Other	user3@example.com	Unknown	Unknown	Unknown
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26-04-2015	0	Not specified	user64@example.com	Bob	Johnson	Bob
EMP1017	Jane Smith	Jane Smith	Finance	Australia	41341	03-05-2015	1	Other	user38@example.com	Jane	Smith	Jane
EMP1018	Alice Brown	Alice Brown	Finance	UK	73602	10-05-2015	0	Not specified	user96@example.com	Alice	Brown	Alice
EMP1019	Eve Adams	Eve Adams	Unknown	India	67531	17-05-2015	5	Female	user151@example.com	Eve	Adams	Eve
EMP1020	Jane Smith	Jane Smith	Hr	USA	68269	24-05-2015	2	Other	user200@example.com	Jane	Smith	Jane
EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015	4	Not specified	user77@example.com	Jane	Smith	Jane
EMP1022	Eve Adams	Eve Adams	Finance	India	38333	07-06-2015	5	Not specified	user128@example.com	Eve	Adams	Eve
EMP1023	Alice Brown	Alice Brown	Marketing	UK	36515	14-06-2015	3	Male	user149@example.com	Alice	Brown	Alice
EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21-06-2015	1	Other	user122@example.com	Sam	Wilson	Sam
EMP1025	Chris Lee	Chris Lee	Unknown	Unknown	52272	28-06-2015	3	Not specified	user41@example.com	Chris	Lee	Chris
EMP1026	Eve Adams	Eve Adams	Sales	Germany	76518	05-07-2015	2	Female	user68@example.com	Eve	Adams	Eve
EMP1027	Alice Brown	Alice Brown	It	UK	41448	12-07-2015	3	Other	user103@example.com	Alice	Brown	Alice
EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015	2	Female	user135@example.com	John	Doe	John

Inconsistent text data was handled using Text-to-Columns, Flash Fill, and TRIM functions. The Name column was split into First Name and Last Name using Text-to-Columns and Flash Fill. TRIM was used to remove extra spaces, ensuring clean and structured data for analysis.

Day 9 – Conditional Formatting for Data Cleaning

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender
EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015	1	Other
EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015	2	Not specified
EMP1003	John Doe	John Doe	Sales	USA	94406	25-01-2015	1	Male
EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015	1	Male
EMP1005	Alice Brown	Alice Brown	It	India	78421	08-02-2015	2	Female
EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	35364	22-02-2015	2	Other
EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015	0	Female
EMP1009	Chris Lee	Chris Lee	Hr	UK	91846	08-03-2015	0	Male
EMP1010	Sam Wilson	Sam Wilson	Finance	UK	64247	15-03-2015	1	Female
EMP1011	Sam Wilson	Sam Wilson	It	USA	91180	22-03-2015	5	Male
EMP1013	Sam Wilson	Sam Wilson	Unknown	India	62158	05-04-2015	4	Not specified
EMP1014	Jane Smith	Jane Smith	Hr	Unknown	59894	12-04-2015	2	Male
EMP1015	Unknown	Unknown	It	India	92443	19-04-2015	3	Other
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26-04-2015	0	Not specified
EMP1017	Jane Smith	Jane Smith	Finance	Australia	41341	03-05-2015	1	Other
EMP1018	Alice Brown	Alice Brown	Finance	UK	73602	10-05-2015	0	Not specified
EMP1019	Eve Adams	Eve Adams	Unknown	India	67531	17-05-2015	5	Female
EMP1020	Jane Smith	Jane Smith	Hr	USA	68269	24-05-2015	2	Other
EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015	4	Not specified
EMP1022	Eve Adams	Eve Adams	Finance	India	38333	07-06-2015	5	Not specified
EMP1023	Alice Brown	Alice Brown	Marketing	UK	36515	14-06-2015	3	Male
EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21-06-2015	1	Other
EMP1025	Chris Lee	Chris Lee	Unknown	Unknown	52272	28-06-2015	3	Not specified
EMP1026	Eve Adams	Eve Adams	Sales	Germany	76518	05-07-2015	2	Female
EMP1027	Alice Brown	Alice Brown	It	UK	41448	12-07-2015	3	Other
EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015	2	Female

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email
EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015	2	Not specified	user176@example.com
EMP1003	John Doe	John Doe	Sales	USA	94406	25-01-2015	1	Male	user168@example.com
EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015	1	Male	user49@example.com
EMP1005	Alice Brown	Alice Brown	It	India	78421	08-02-2015	2	Female	user24@example.com
EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	35364	22-02-2015	2	Other	user26@example.com
EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015	0	Female	user25@example.com
EMP1009	Chris Lee	Chris Lee	Hr	UK	91846	08-03-2015	0	Male	user153@example.com
EMP1010	Sam Wilson	Sam Wilson	Finance	UK	64247	15-03-2015	1	Female	user18@example.com
EMP1011	Sam Wilson	Sam Wilson	It	USA	91180	22-03-2015	5	Male	user1@example.com
EMP1013	Sam Wilson	Sam Wilson	Unknown	India	62158	05-04-2015	4	Not specified	user180@example.com
EMP1014	Jane Smith	Jane Smith	Hr	Unknown	59894	12-04-2015	2	Male	user184@example.com
EMP1015	Unknown	Unknown	It	India	92443	19-04-2015	3	Other	user3@example.com
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26-04-2015	0	Not specified	user64@example.com
EMP1017	Jane Smith	Jane Smith	Finance	Australia	41341	03-05-2015	1	Other	user38@example.com
EMP1018	Alice Brown	Alice Brown	Finance	UK	73602	10-05-2015	0	Not specified	user96@example.com
EMP1019	Eve Adams	Eve Adams	Unknown	India	67531	17-05-2015	5	Female	user151@example.com
EMP1020	Jane Smith	Jane Smith	Hr	USA	68269	24-05-2015	2	Other	user200@example.com
EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015	4	Not specified	user77@example.com
EMP1022	Eve Adams	Eve Adams	Finance	India	38333	07-06-2015	5	Not specified	user128@example.com
EMP1023	Alice Brown	Alice Brown	Marketing	UK	36515	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21-06-2015	1	Other	user122@example.com
EMP1025	Chris Lee	Chris Lee	Unknown	Unknown	52272	28-06-2015	3	Not specified	user41@example.com
EMP1026	Eve Adams	Eve Adams	Sales	Germany	76518	05-07-2015	2	Female	user68@example.com
EMP1027	Alice Brown	Alice Brown	It	UK	41448	12-07-2015	3	Other	user103@example.com
EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015	2	Female	user135@example.com

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email
EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015	2	Not specified	user176@example.com
EMP1003	John Doe	John Doe	Sales	USA	94406	25-01-2015	1	Male	user168@example.com
EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015	1	Male	user49@example.com
EMP1005	Alice Brown	Alice Brown	It	India	78421	08-02-2015	2	Female	user24@example.com
EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	35364	22-02-2015	2	Other	user26@example.com
EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015	0	Female	user25@example.com
EMP1009	Chris Lee	Chris Lee	Hr	UK	91846	08-03-2015	0	Male	user153@example.com
EMP1010	Sam Wilson	Sam Wilson	Finance	UK	64247	15-03-2015	1	Female	user18@example.com
EMP1011	Sam Wilson	Sam Wilson	It	USA	91180	22-03-2015	5	Male	user1@example.com
EMP1013	Sam Wilson	Sam Wilson	Unknown	India	62158	05-04-2015	4	Not specified	user180@example.com
EMP1014	Jane Smith	Jane Smith	Hr	Unknown	59894	12-04-2015	2	Male	user184@example.com
EMP1015	Unknown	Unknown	It	India	92443	19-04-2015	3	Other	user3@example.com
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26-04-2015	0	Not specified	user64@example.com
EMP1017	Jane Smith	Jane Smith	Finance	Australia	41341	03-05-2015	1	Other	user38@example.com
EMP1018	Alice Brown	Alice Brown	Finance	UK	73602	10-05-2015	0	Not specified	user96@example.com
EMP1019	Eve Adams	Eve Adams	Unknown	India	67531	17-05-2015	5	Female	user151@example.com
EMP1020	Jane Smith	Jane Smith	Hr	USA	68269	24-05-2015	2	Other	user200@example.com
EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015	4	Not specified	user77@example.com
EMP1022	Eve Adams	Eve Adams	Finance	India	38333	07-06-2015	5	Not specified	user128@example.com
EMP1023	Alice Brown	Alice Brown	Marketing	UK	36515	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21-06-2015	1	Other	user122@example.com
EMP1025	Chris Lee	Chris Lee	Unknown	Unknown	52272	28-06-2015	3	Not specified	user41@example.com
EMP1026	Eve Adams	Eve Adams	Sales	Germany	76518	05-07-2015	2	Female	user68@example.com
EMP1027	Alice Brown	Alice Brown	It	UK	41448	12-07-2015	3	Other	user103@example.com
EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015	2	Female	user135@example.com

FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelpPower Pivot

Cut

Copy

Format Painter

Calibri

11

A⁺

A⁻

B

I

U

□

□

□

□

□

□

Wrap Text

Merge & Center

General

□

□

□

□

□

□

□

□

□

□

□

□

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

AutoSum

Fill

Clear

Sort & Filter

Find & Select

Add-ins

Comments

Share

Font

Alignment

Number

Styles

Cells

Editing

Add-ins

K1

▼

✕

✕

✕

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email							
1	EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015		1 Other	user106@example.com							
2	EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015		2 Not specified	user176@example.com							
3	EMP1003	John Doe	John Doe	Sales	USA	94406	25-01-2015		1 Male	user168@example.com							
4	EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015		1 Male	user49@example.com							
5	EMP1005	Alice Brown	Alice Brown	It	India	78421	08-02-2015		2 Female	user24@example.com							
6	EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	35364	22-02-2015		2 Other	user26@example.com							
7	EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015		0 Female	user25@example.com							
8	EMP1009	Chris Lee	Chris Lee	Hr	UK	91846	08-03-2015		0 Male	user153@example.com							
9	EMP1010	Sam Wilson	Sam Wilson	Finance	UK	64247	15-03-2015		1 Female	user18@example.com							
10	EMP1011	Sam Wilson	Sam Wilson	It	USA	91180	22-03-2015		5 Male	user1@example.com							
11	EMP1013	Sam Wilson	Sam Wilson	Unknown	India	62158	05-04-2015		4 Not specified	user180@example.com							
12	EMP1014	Jane Smith	Jane Smith	Unknown	Hr	59804	12-04-2015		2 Male	user194@example.com							
13	EMP1015	Unknown	Unknown	It	India	92443	19-04-2015		3 Other	user3@example.com							
14	EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26-04-2015		0 Not specified	user64@example.com							
15	EMP1017	Jane Smith	Jane Smith	Finance	Australia	41341	03-05-2015		1 Other	user38@example.com							
16	EMP1018	Alice Brown	Alice Brown	Finance	UK	73602	10-05-2015		0 Not specified	user96@example.com							
17	EMP1019	Eve Adams	Eve Adams	Unknown	India	67531	17-05-2015		5 Female	user151@example.com							
18	EMP1020	Jane Smith	Jane Smith	Hr	USA	68269	24-05-2015		2 Other	user200@example.com							
19	EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015		4 Not specified	user77@example.com							
20	EMP1022	Eve Adams	Eve Adams	Finance	India	38333	07-06-2015		5 Not specified	user128@example.com							
21	EMP1023	Alice Brown	Alice Brown	Marketing	UK	36515	14-06-2015		3 Male	user149@example.com							
22	EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21-06-2015		1 Other	user112@example.com							
23	EMP1025	Chris Lee	Chris Lee	Unknown	Germany	527172	28-06-2015		3 Not specified	user41@example.com							
24	EMP1026	Eve Adams	Eve Adams	Sales	Germany	75518	05-07-2015		2 Female	user68@example.com							
25	EMP1027	Alice Brown	Alice Brown	It	UK	41448	12-07-2015		3 Other	user103@example.com							
26	EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015		2 Female	user135@example.com							

<

...

DAY6-DATACLEANING

Day7-Power query

DAY8-Formatting

Day9-Data cleaning

+

◀

▶

Conditional formatting was applied to visually identify data issues. Missing values were highlighted using a custom rule and outliers such as high were highlighted using conditional rules. This helped in quickly identifying and analysing inconsistencies in the dataset.

Day 10 – Descriptive Statistics

File Home Insert Draw Layout Formulas Data Review View Help Power Pivot

Get Data: From Text/CSV, Recent Sources, From Web, Existing Connections, From Table/Range

Data: Refresh All, Properties, Workbook Links, Queries & Connections

Sort & Filter: Sort, Filter, Advanced

Data Tools: Flash Fill, Remove Duplicates, Consolidate, Data Model, Text to Columns, Data Validation, What-If Analysis, Forecast Sheet

Grouping: Group, Ungroup, Subtotal, Outline, Analysis

Formulas: L18, fx, Paste, Paste Options

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	EmployeeID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email							
1	EMP1001	Eve Adams	Eve Adams	Hr	Germany	94265	15-01-2015		1 Other	user106@example.com							
2	EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015		2 Not specified	user176@example.com							
3	EMP1003	John Doe	John Doe	Sales	USA	94406	25-01-2015		1 Male	user168@example.com							
4	EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015		1 Male	user49@example.com							
5	EMP1005	Alice Brown	Alice Brown	It	India	78421	08-02-2015		2 Female	user24@example.com							
6	EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	35364	22-02-2015		2 Other	user26@example.com							
7	EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015		0 Female	user25@example.com							
8	EMP1009	Chris Lee	Chris Lee	Hr						user153@example.com							
9	EMP1010	Sam Wilson	Sam Wilson	Finance						user18@example.com							
10	EMP1011	Sam Wilson	Sam Wilson	It						user1@example.com							
11	EMP1013	Sam Wilson	Sam Wilson	Unknown						user180@example.com							
12	EMP1014	Jane Smith	Jane Smith	Hr						user184@example.com							
13	EMP1015	Unknown	Unknown	It						user3@example.com							
14	EMP1016	Bob Johnson	Bob Johnson	Unknown						user64@example.com							
15	EMP1017	Jane Smith	Jane Smith	Finance						user98@example.com							
16	EMP1018	Alice Brown	Alice Brown	Finance						user96@example.com							
17	EMP1019	Eve Adams	Eve Adams	Unknown						user151@example.com							
18	EMP1020	Jane Smith	Jane Smith	Hr						user200@example.com							
19	EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015		4 Not specified	user77@example.com							
20	EMP1022	Eve Adams	Eve Adams	Finance	India	38333	07-06-2015		5 Not specified	user128@example.com							
21	EMP1023	Alice Brown	Alice Brown	Marketing	UK	36515	14-06-2015		3 Male	user149@example.com							
22	EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21-06-2015		1 Other	user122@example.com							
23	EMP1025	Chris Lee	Chris Lee	Unknown		52272	28-06-2015		3 Not specified	user41@example.com							
24	EMP1026	Eve Adams	Eve Adams	Sales	Germany	76518	05-07-2015		2 Female	user68@example.com							
25	EMP1027	Alice Brown	Alice Brown	It	UK	41448	12-07-2015		3 Other	user103@example.com							
26	EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015		2 Female	user135@example.com							

Ready Accessibility: Investigate

Day 11 – Filtering, Sorting & Exploring Data

Excel ribbon: File, Home, Insert, Draw, Page Layout, Formulas, Data, Review, View, Help, Power Pivot, Table Design, Query. Data Tools: Get & Transform Data, Queries & Connections, Sort & Filter, Data Tools, Forecast, Analysis.

Table: Employee Data

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email
EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015	2	Not specified	user176@example.com
EMP1003	John Doe	John Doe	Finance	USA	94406	25-01-2015	1	Male	user168@example.com
EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015	1	Male	user49@example.com
EMP1005	Alice Brown	Alice Brown	It	India	78421	08-02-2015	2	Female	user24@example.com
EMP1007	Chris Lee	Chris Lee	Unknown	Unknown	35364	22-02-2015	2	Other	user26@example.com
EMP1008	Alice Brown	Alice Brown	Unknown	Germany	97881	01-03-2015	0	Female	user25@example.com
EMP1009	Chris Lee	Chris Lee	Unknown	UK	91846	08-03-2015	0	Male	user153@example.com
EMP1010	Sam Wilson	Sam Wilson	Unknown	UK	64247	15-03-2015	1	Female	user18@example.com
EMP1011	Sam Wilson	Sam Wilson	Unknown	USA	91180	22-03-2015	5	Male	user1@example.com
EMP1013	Sam Wilson	Sam Wilson	Unknown	India	62158	05-04-2015	4	Not specified	user180@example.com
EMP1014	Jane Smith	Jane Smith	Unknown	Unknown	59894	12-04-2015	2	Male	user184@example.com
EMP1015	Unknown	Unknown	Unknown	India	92443	19-04-2015	3	Other	user3@example.com
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26-04-2015	0	Not specified	user64@example.com
EMP1017	Jane Smith	Jane Smith	Unknown	Australia	41341	03-05-2015	1	Other	user38@example.com
EMP1018	Alice Brown	Alice Brown	Unknown	UK	73602	10-05-2015	0	Not specified	user96@example.com
EMP1019	Eve Adams	Eve Adams	Unknown	India	67531	17-05-2015	5	Female	user151@example.com
EMP1020	Jane Smith	Jane Smith	Unknown	USA	68269	24-05-2015	2	Other	user200@example.com
EMP1021	Jane Smith	Jane Smith	Unknown	Unknown	98922	31-05-2015	4	Not specified	user77@example.com
EMP1022	Eve Adams	Eve Adams	Unknown	India	38333	07-06-2015	5	Not specified	user128@example.com
EMP1023	Alice Brown	Alice Brown	Unknown	UK	36515	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Sam Wilson	Unknown	Unknown	91005	21-06-2015	1	Other	user122@example.com
EMP1026	Chris Lee	Chris Lee	Unknown	Unknown	52272	28-06-2015	3	Not specified	user41@example.com
EMP1027	Eve Adams	Eve Adams	Unknown	Germany	76518	05-07-2015	2	Female	user68@example.com
EMP1028	John Doe	John Doe	Unknown	UK	41448	12-07-2015	3	Other	user103@example.com
EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015	2	Female	user135@example.com

Table: Employee Data (Filtered by Country: Germany)

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email
EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015	1	Other	user106@example.com
EMP1009	Chris Lee	Chris Lee	Hr	UK	91846	08-03-2015	0	Male	user153@example.com
EMP1014	Jane Smith	Jane Smith	Hr	Unknown	59894	12-04-2015	2	Male	user184@example.com
EMP1020	Jane Smith	Jane Smith	Hr	USA	68269	24-05-2015	2	Other	user200@example.com
EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015	4	Not specified	user77@example.com
EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015	2	Female	user135@example.com
EMP1031	Eve Adams	Eve Adams	Hr	Germany	92299	09-08-2015	3	Male	user143@example.com
EMP1032	Sam Wilson	Sam Wilson	Hr	Unknown	95586	16-08-2015	3	Other	user190@example.com
EMP1037	Alice Brown	Alice Brown	Hr	Unknown	64796	20-09-2015	0	Other	user98@example.com
EMP1039	Chris Lee	Chris Lee	Hr	UK	48224	04-10-2015	3	Not specified	user42@example.com
EMP1047	Chris Lee	Chris Lee	Hr	India	58245	29-11-2015	4	Not specified	user43@example.com
EMP1063	Eve Adams	Eve Adams	Hr	USA	36541	20-03-2016	2	Other	user97@example.com
EMP1097	Sam Wilson	Sam Wilson	Hr	Germany	89333	13-11-2016	1	Male	user113@example.com
EMP1102	Alice Brown	Alice Brown	Hr	Germany	42096	18-12-2016	0	Female	user54@example.com
EMP1108	Sam Wilson	Sam Wilson	Hr	Unknown	66761	29-01-2017	1	Not specified	user93@example.com
EMP1132	Eve Adams	Eve Adams	Hr	USA	38678	16-07-2017	3	Other	user188@example.com
EMP1137	Chris Lee	Chris Lee	Hr	Germany	84968	20-08-2017	0	Not specified	user165@example.com

Table: Employee Data (Filtered by Country: Germany)

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email
EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015	2	Not specified	user176@example.com
EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015	1	Male	user49@example.com
EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015	0	Female	user25@example.com
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26-04-2015	0	Not specified	user64@example.com
EMP1026	Eve Adams	Eve Adams	Sales	Germany	76518	05-07-2015	2	Female	user68@example.com
EMP1031	Eve Adams	Eve Adams	Hr	Germany	92299	09-08-2015	3	Male	user143@example.com
EMP1041	Alice Brown	Alice Brown	It	Germany	33956	18-10-2015	0	Not specified	user47@example.com
EMP1046	Chris Lee	Chris Lee	It	Germany	92701	22-11-2015	3	Other	user199@example.com
EMP1060	Unknown	Unknown	Marketing	Germany	68269.25		5	Unknown	user112@example.com
EMP1097	Sam Wilson	Sam Wilson	Hr	Germany	89333	13-11-2016	1	Male	user113@example.com
EMP1102	Alice Brown	Alice Brown	Hr	Germany	42096	18-12-2016	0	Female	user54@example.com
EMP1122	Bob Johnson	Bob Johnson	Sales	Germany	52036	07-05-2017	3	Male	user171@example.com
EMP1126	Bob Johnson	Bob Johnson	Unknown	Germany	71494	04-06-2017	3	Male	user100@example.com
EMP1137	Chris Lee	Chris Lee	Hr	Germany	84968	20-08-2017	0	Not specified	user165@example.com

Filter:Country-Germany

File Home Insert Draw Page Layout Formulas Data Review View Help Power Pivot Table Design Query

Get Data From Text/CSV Recent Sources From Web Existing Connections From Table/Range

Queries & Connections Properties Workbook Links Refresh All

Sort Filter Clear Reapply Advanced

Flash Fill Consolidate Remove Duplicates Data Model What-If Analysis Forecast Sheet Subtotal Data Analysis

Comments Share

D2

Employee ID Name PROPER NAMES Department Country Salary Date of Joining Performance Rating Gender Email

Sort Smallest to Largest

Sort Largest to Smallest

Sort by Color

Sheet View

Clear Filter From "Salary"

Filter by Color

Number Filters

Search

(Select All)

30286

30402

30686

31293

33264

33468

33483

33956

OK Cancel

DAY8-Formatting DAY9-Data cleaning DAY10-Descriptive statistics DAY11-Filtering/Sorting

File Home Insert Draw Page Layout Formulas Data Review View Help Power Pivot Table Design Query

Get Data From Text/CSV Recent Sources From Web Existing Connections From Table/Range

Queries & Connections Properties Workbook Links Refresh All

Sort Filter Clear Reapply Advanced

Flash Fill Consolidate Remove Duplicates Data Model What-If Analysis Forecast Sheet Subtotal Data Analysis

Comments Share

D2

Employee ID Name PROPER NAMES Department Country Salary Date of Joining Performance Rating Gender Email

EMP1083 Sam Wilson Sam Wilson Sales India 99621 07-08-2016 0 Other user145@example.com

EMP1021 Jane Smith Jane Smith Hr Unknown 98922 31-05-2015 4 Not specified user77@example.com

EMP1149 Sam Wilson Sam Wilson Marketing India 98475 12-11-2017 5 Other user30@example.com

EMP1062 Chris Lee Chris Lee It UK 97981 13-03-2016 5 Female user71@example.com

EMP1008 Alice Brown Alice Brown It Germany 97881 01-03-2015 0 Female user25@example.com

EMP1066 Sam Wilson Sam Wilson Unknown Australia 96365 10-04-2016 3 Other user90@example.com

EMP1075 Unknown Unknown Sales India 96142 12-06-2016 1 Other user17@example.com

EMP1095 Alice Brown Alice Brown Finance Australia 96101 30-10-2016 0 Other user155@example.com

EMP1081 Jane Smith Jane Smith Sales USA 95621 24-07-2016 3 Female user156@example.com

EMP1032 Sam Wilson Sam Wilson Hr Unknown 95586 16-08-2015 3 Other user190@example.com

EMP1033 Chris Lee Chris Lee Unknown USA 95367 23-08-2015 0 Male user136@example.com

EMP1118 Chris Lee Chris Lee Sales India 95032 09-04-2017 3 Not specified user23@example.com

EMP1003 John Doe John Doe Sales USA 94406 25-01-2015 1 Male user168@example.com

EMP1144 Eve Adams Eve Adams Unknown Unknown 94148 08-10-2017 2 Not specified user53@example.com

EMP1029 Bob Johnson Bob Johnson Marketing UK 92959 26-07-2015 5 Not specified user83@example.com

EMP1046 Chris Lee Chris Lee It Germany 92701 22-11-2015 3 Other user199@example.com

EMP1015 Unknown Unknown It India 92443 19-04-2015 3 Other user3@example.com

EMP1031 Eve Adams Eve Adams Hr Germany 92299 09-08-2015 3 Male user143@example.com

EMP1030 Unknown Unknown Marketing Unknown 92259 09-08-2015 1 Unknown user109@example.com

EMP1009 Chris Lee Chris Lee Hr UK 91846 08-03-2015 0 Male user153@example.com

EMP1143 Chris Lee Chris Lee Sales UK 91620 01-10-2017 0 Not specified user114@example.com

EMP1011 Sam Wilson Sam Wilson It USA 91180 22-03-2015 5 Male user1@example.com

EMP1024 Sam Wilson Sam Wilson Finance Unknown 91005 21-06-2015 1 Other user122@example.com

EMP1016 Bob Johnson Bob Johnson Unknown Germany 90271 26-04-2015 0 Not specified user64@example.com

EMP1125 John Doe John Doe Finance 90019 28-05-2017 5 Not specified user52@example.com

EMP1036 Alice Brown Alice Brown It USA 89781 13-09-2015 0 Female user31@example.com

DAY8-Formatting DAY9-Data cleaning DAY10-Descriptive statistics DAY11-Filtering/Sorting

File Home Insert Draw Page Layout Formulas Data Review View Help Power Pivot Table Design Query

Get Data From Text/CSV Recent Sources From Web Existing Connections From Table/Range

Queries & Connections Properties Workbook Links Refresh All

Sort Filter Clear Reapply Advanced

Flash Fill Consolidate Remove Duplicates Data Model What-If Analysis Forecast Sheet Subtotal Data Analysis

Comments Share

D2

Employee ID Name PROPER NAMES Department Country Salary Date of Joining Performance Rating Gender Email

EMP1083 Sam Wilson Sam Wilson Sales India 99621 07-08-2016 0 Other user145@example.com

EMP1021 Jane Smith Jane Smith Hr Unknown 98922 31-05-2015 4 Not specified user77@example.com

EMP1149 Sam Wilson Sam Wilson Marketing India 98475 12-11-2017 5 Other user30@example.com

EMP1062 Chris Lee Chris Lee It UK 97981 13-03-2016 5 Female user71@example.com

EMP1008 Alice Brown Alice Brown It Germany 97881 01-03-2015 0 Female user25@example.com

EMP1066 Sam Wilson Sam Wilson Unknown Australia 96365 10-04-2016 3 Other user90@example.com

EMP1075 Unknown Unknown Sales India 96142 12-06-2016 1 Other user17@example.com

EMP1095 Alice Brown Alice Brown Finance Australia 96101 30-10-2016 0 Other user155@example.com

EMP1081 Jane Smith Jane Smith Sales USA 95621 24-07-2016 3 Female user156@example.com

EMP1032 Sam Wilson Sam Wilson Hr Unknown 95586 16-08-2015 3 Other user190@example.com

EMP1033 Chris Lee Chris Lee Unknown USA 95367 23-08-2015 0 Male user136@example.com

EMP1118 Chris Lee Chris Lee Sales India 95032 09-04-2017 3 Not specified user23@example.com

EMP1003 John Doe John Doe Sales USA 94406 25-01-2015 1 Male user168@example.com

EMP1144 Eve Adams Eve Adams Unknown Unknown 94148 08-10-2017 2 Not specified user53@example.com

EMP1029 Bob Johnson Bob Johnson Marketing UK 92959 26-07-2015 5 Not specified user83@example.com

EMP1046 Chris Lee Chris Lee It Germany 92701 22-11-2015 3 Other user199@example.com

EMP1015 Unknown Unknown It India 92443 19-04-2015 3 Other user3@example.com

EMP1031 Eve Adams Eve Adams Hr Germany 92299 09-08-2015 3 Male user143@example.com

EMP1030 Unknown Unknown Marketing Unknown 92259 09-08-2015 1 Unknown user109@example.com

EMP1009 Chris Lee Chris Lee Hr UK 91846 08-03-2015 0 Male user153@example.com

EMP1143 Chris Lee Chris Lee Sales UK 91620 01-10-2017 0 Not specified user114@example.com

EMP1011 Sam Wilson Sam Wilson It USA 91180 22-03-2015 5 Male user1@example.com

EMP1024 Sam Wilson Sam Wilson Finance Unknown 91005 21-06-2015 1 Other user122@example.com

EMP1016 Bob Johnson Bob Johnson Unknown Germany 90271 26-04-2015 0 Not specified user64@example.com

EMP1125 John Doe John Doe Finance 90019 28-05-2017 5 Not specified user52@example.com

EMP1036 Alice Brown Alice Brown It USA 89781 13-09-2015 0 Female user31@example.com

Top 10 AutoFilter

Show

Top 5 Items

OK Cancel

DAY8-Formatting DAY9-Data cleaning DAY10-Descriptive statistics DAY11-Filtering/Sorting

File Home Insert Draw Page Layout Formulas Data Review View Help Power Pivot Table Design Query										
Get From Text/CSV Recent Sources Existing Connections From Web From Table/Range Refresh All Queries & Connections Properties Workbook Links Sort Filter Clear Reapply Advanced Flash Fill Remove Duplicates Consolidate Data Model Text to Columns Data Validation What-If Analysis Forecast Sheet Subtotal Outline Group Ungroup Data Analysis										
D2										
A	B	C	D	E	F	G	H	I	J	
Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email	
EMP1083	Sam Wilson	Sam Wilson	Sales	India	99621	07-08-2016	0	Other	user145@example.com	
EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015	4	Not specified	user77@example.com	
EMP1149	Sam Wilson	Sam Wilson	Marketing	India	98475	12-11-2017	5	Other	user30@example.com	
EMP1062	Chris Lee	Chris Lee	It	UK	97981	13-03-2016	5	Female	user71@example.com	
EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015	0	Female	user25@example.com	

File Home Insert Draw Page Layout Formulas Data Review View Help Power Pivot Table Design Query										
Get From Text/CSV Recent Sources Existing Connections From Web From Table/Range Refresh All Queries & Connections Properties Workbook Links Sort Filter Clear Reapply Advanced Flash Fill Remove Duplicates Consolidate Data Model Text to Columns Data Validation What-If Analysis Forecast Sheet Subtotal Outline Group Ungroup Data Analysis										
D2										
A	B	C	D	E	F	G	H	I	J	
Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email	
EMP1083	Sam Wilson	Sam Wilson	Sales	India	99621	07-08-2016	0	Other	user145@example.com	
EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015	4	Not specified	user77@example.com	
EMP1149	Sam Wilson	Sam Wilson	Marketing	India	98475	12-11-2017	5	Other	user30@example.com	
EMP1062	Chris Lee	Chris Lee	It	UK	97981	13-03-2016	5	Female	user71@example.com	
EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015	0	Female	user25@example.com	
EMP1066	Sam Wilson	Sam Wilson	Unknown	Australia	96365	10-04-2016	3	Other	user90@example.com	
EMP1075	Unknown	Unknown	Sales	India	96142	12-06-2016	1	Other	user17@example.com	
EMP1095	Alice Brown	Alice Brown	Finance	Australia	96101	30-10-2016	0	Other	user155@example.com	
EMP1081	Jane Smith	Jane Smith	Sales	USA	95621	24-07-2016	3	Female	user156@example.com	
EMP1032	Sam Wilson	Sam Wilson	Hr	Unknown			3	Other	user190@example.com	
EMP1033	Chris Lee	Chris Lee	Unknown	USA			0	Male	user136@example.com	
EMP1118	Chris Lee	Chris Lee	Sales	India			3	Not specified	user23@example.com	
EMP1003	John Doe	John Doe	Sales	USA			1	Male	user168@example.com	
EMP1144	Eve Adams	Eve Adams	Unknown	Unknown			2	Not specified	user53@example.com	
EMP1029	Bob Johnson	Bob Johnson	Marketing	UK			5	Not specified	user83@example.com	
EMP1046	Chris Lee	Chris Lee	It	Germany			3	Other	user199@example.com	
EMP1015	Unknown	Unknown	It	India	92443	19-04-2015	3	Other	user3@example.com	
EMP1031	Eve Adams	Eve Adams	Hr	Germany	92299	09-08-2015	3	Male	user143@example.com	
EMP1030	Unknown	Unknown	Marketing	Unknown	92259		1	Unknown	user109@example.com	
EMP1009	Chris Lee	Chris Lee	Hr	UK	91846	08-03-2015	0	Male	user153@example.com	
EMP1143	Chris Lee	Chris Lee	Sales	UK	91620	01-10-2017	0	Not specified	user114@example.com	
EMP1011	Sam Wilson	Sam Wilson	It	USA	91180	22-03-2015	5	Male	user1@example.com	
EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21-06-2015	1	Other	user122@example.com	
EMP1016	Bob Johnson	Bob Johnson	Finance	Germany	90271	26-04-2015	0	Not specified	user64@example.com	
EMP1125	John Doe	John Doe	Finance	Unknown	90019	28-05-2017	5	Not specified	user52@example.com	
EMP1036	Alice Brown	Alice Brown	It	USA	89781	13-09-2015	0	Female	user31@example.com	

File Home Insert Draw Page Layout Formulas Data Review View Help Power Pivot Table Design Query										
Get From Text/CSV Recent Sources Existing Connections From Web From Table/Range Refresh All Queries & Connections Properties Workbook Links Sort Filter Clear Reapply Advanced Flash Fill Remove Duplicates Consolidate Data Model Text to Columns Data Validation What-If Analysis Forecast Sheet Subtotal Outline Group Ungroup Data Analysis										
D2										
A	B	C	D	E	F	G	H	I	J	
Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email	
EMP1094	Bob Johnson	Bob Johnson	It	India	33264	23-10-2016	0	Female	user148@example.com	
EMP1110	Chris Lee	Chris Lee	Sales	India	31293		2	Female	user44@example.com	
EMP1050	Chris Lee	Chris Lee	Unknown		30686		2	Female	user95@example.com	
EMP1128	Chris Lee	Chris Lee	Sales	USA	30402	18-06-2017	1	Not specified	user167@example.com	
EMP1052	Eve Adams	Eve Adams	Marketing	Unknown	30280	03-01-2016	1	Not specified	user152@example.com	

Filtering and sorting techniques were applied to explore the dataset. Multiple filters were used to analyze specific groups based on department and country. The Salary column was sorted to identify the top 5 and bottom 5 earning employees.

It was observed that higher salaries were concentrated in certain departments. Filtering helped in understanding patterns across different categories.

Day 12 – Date & Time Functions

The screenshot displays an Excel spreadsheet with a dataset of employee information and a table of date/time functions. The dataset includes columns for Employee ID, Name, Proper Names, Department, Country, Salary, Date of Joining, Performance Rating, Gender, and Email. The date/time functions table includes Today, Now, Year, Month, Day, WeekDay, and Datedif.

Employee ID	Name	PROPER NAMES	Department	Country	Salary	Date of Joining	Performance Rating	Gender	Email
EMP1001	Eve Adams	Eve Adams	Hr	Germany	54255	11-01-2015	1	Other	user106@example.com
EMP1002	Alice Brown	Alice Brown	Finance	Germany	63184	18-01-2015	2	Not specified	user176@example.com
EMP1003	John Doe	John Doe	Sales	USA	94406	25-01-2015	1	Male	user168@example.com
EMP1004	Jane Smith	Jane Smith	Sales	Germany	64120	01-02-2015	1	Male	user49@example.com
EMP1005	Alice Brown	Alice Brown	It	India	78421	08-02-2015	2	Female	user24@example.com
EMP1007	Chris Lee	Chris Lee	Marketing	Unknown	35364	22-02-2015	2	Other	user26@example.com
EMP1008	Alice Brown	Alice Brown	It	Germany	97881	01-03-2015	0	Female	user25@example.com
EMP1009	Chris Lee	Chris Lee	Hr	UK	91846	08-03-2015	0	Male	user153@example.com
EMP1010	Sam Wilson	Sam Wilson	Finance	UK	64247	15-03-2015	1	Female	user18@example.com
EMP1011	Sam Wilson	Sam Wilson	It	USA	91180	22-03-2015	5	Male	user1@example.com
EMP1013	Sam Wilson	Sam Wilson	Unknown	India	62158	05-04-2015	4	Not specified	user180@example.com
EMP1014	Jane Smith	Jane Smith	Hr	Unknown	59894	12-04-2015	2	Male	user184@example.com
EMP1015	Unknown	Unknown	It	India	92443	19-04-2015	3	Other	user3@example.com
EMP1016	Bob Johnson	Bob Johnson	Unknown	Germany	90271	26-04-2015	0	Not specified	user64@example.com
EMP1017	Jane Smith	Jane Smith	Finance	Australia	41341	03-05-2015	1	Other	user38@example.com
EMP1018	Alice Brown	Alice Brown	Finance	UK	73602	10-05-2015	0	Not specified	user96@example.com
EMP1019	Eve Adams	Eve Adams	Unknown	India	67531	17-05-2015	5	Female	user131@example.com
EMP1020	Jane Smith	Jane Smith	Hr	USA	68269	24-05-2015	2	Other	user200@example.com
EMP1021	Jane Smith	Jane Smith	Hr	Unknown	98922	31-05-2015	4	Not specified	user77@example.com
EMP1022	Eve Adams	Eve Adams	Finance	India	38333	07-06-2015	5	Not specified	user128@example.com
EMP1023	Alice Brown	Alice Brown	Marketing	UK	36515	14-06-2015	3	Male	user149@example.com
EMP1024	Sam Wilson	Sam Wilson	Finance	Unknown	91005	21-06-2015	1	Other	user122@example.com
EMP1025	Chris Lee	Chris Lee	Unknown	Unknown	52272	28-06-2015	3	Not specified	user41@example.com
EMP1026	Eve Adams	Eve Adams	Sales	Germany	76518	05-07-2015	2	Female	user68@example.com
EMP1027	Alice Brown	Alice Brown	It	UK	41448	12-07-2015	3	Other	user103@example.com
EMP1028	John Doe	John Doe	Hr	UK	49282	19-07-2015	2	Female	user135@example.com

Date and Time functions	
Today	07-05-2026
Now	07-05-2026 10:51
Year	2015
Month	1
Day	11
WeekDay	1
Datedif	11

Date and time functions such as TODAY(), NOW(), YEAR(), MONTH(), WEEKDAY(), and DATEDIF() were used to analyze date-related data. The Date of Joining column was used to calculate employee experience in years using the DATEDIF function.

Day 13 – Lookup & Reference Functions (Part 1)

Use: VLOOKUP()

The screenshot displays an Excel spreadsheet with a dataset of employee information. The dataset includes columns for Employee ID, Name, Salary, and Department. The VLOOKUP function is used to retrieve employee details based on Employee ID.

Employee ID	Name	Salary	Department
EMP1001	Eve Adams	54255	Hr
EMP1002	Alice Brown	63184	Finance
EMP1003	John Doe	94406	Sales
EMP1004	Jane Smith	64120	Sales
EMP1005	Alice Brown	78421	It
EMP1007	Chris Lee	35364	Marketing
EMP1008	Alice Brown	97881	It
EMP1009	Chris Lee	91846	Hr
EMP1010	Sam Wilson	64247	Finance
EMP1011	Sam Wilson	91180	It
EMP1013	Sam Wilson	62158	Unknown
EMP1014	Jane Smith	59894	Hr
EMP1015	Unknown	92443	It
EMP1016	Bob Johnson	90271	Unknown
EMP1017	Jane Smith	41341	Finance
EMP1018	Alice Brown	73602	Finance
EMP1019	Eve Adams	67531	Unknown
EMP1020	Jane Smith	68269.25	Hr
EMP1021	Jane Smith	98922	Hr
EMP1022	Eve Adams	38333	Finance
EMP1023	Alice Brown	36515	Marketing
EMP1024	Sam Wilson	91005	Finance
EMP1025	Chris Lee	52272	Unknown
EMP1026	Eve Adams	76518	Sales
EMP1027	Alice Brown	41448	It
EMP1028	John Doe	49282	Hr

The VLOOKUP() function was used to duplicate the dataset by retrieving employee details from the original table based on Employee ID. Different columns such as Name, Department, and Salary were fetched using column index numbers and exact match lookup. This helped in understanding lookup and reference operations in Excel.

Day 14 – INDEX(), MATCH()

Formula Bar: `=INDEX('Day10-Descriptive statistics'!F:F,MATCH([@Employee ID],'Day10-Descriptive statistics'!A:A,0))`

Employee ID	Name	Department	Salary
EMP1001	Eve Adams	Hr	54255
EMP1002	Alice Brown	Finance	63184
EMP1003	John Doe	Sales	94406
EMP1004	Jane Smith	Sales	64120
EMP1005	Alice Brown	It	78423
EMP1007	Chris Lee	Marketing	35364
EMP1008	Alice Brown	It	97881
EMP1009	Chris Lee	Hr	91846
EMP1010	Sam Wilson	Finance	64247
EMP1011	Sam Wilson	It	91180
EMP1013	Sam Wilson	Unknown	62158
EMP1014	Jane Smith	Hr	59894
EMP1015	Unknown	It	92443
EMP1016	Bob Johnson	Unknown	90271
EMP1017	Jane Smith	Finance	41341
EMP1018	Alice Brown	Finance	73602
EMP1019	Eve Adams	Unknown	67531
EMP1020	Jane Smith	Hr	68269.25
EMP1021	Jane Smith	Hr	98922
EMP1022	Eve Adams	Finance	38333
EMP1023	Alice Brown	Marketing	36515
EMP1024	Sam Wilson	Finance	91005
EMP1025	Chris Lee	Unknown	52272
EMP1026	Eve Adams	Sales	76518
EMP1027	Alice Brown	It	41448

The INDEX() and MATCH() functions were used together to retrieve related data from another table based on Employee ID. MATCH() was used to locate the position of the Employee ID in the lookup table, and INDEX() returned the corresponding value from the required column such as Salary. This method helped in understanding advanced lookup techniques and is more flexible than VLOOKUP() because it can search in any direction.

Day 15 – Pivot Tables (Part 1)

PivotTable Fields Task Pane:

- Average salary by Department:** Row Labels - Average of Salary
- Employee count by Country:** Row Labels - Count of Employee ID
- Performance Rating by Gender:** Row Labels - Sum of Performance Rating

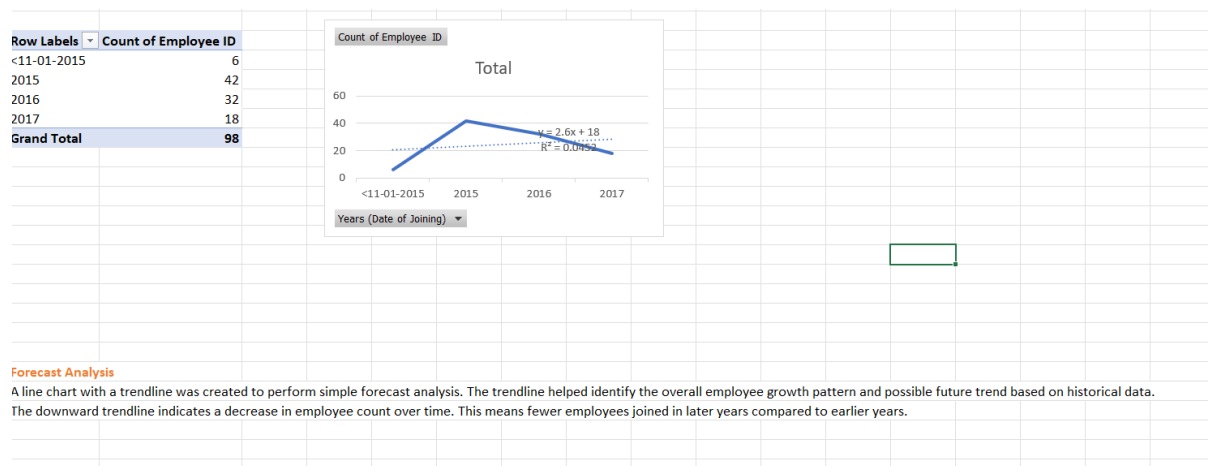
Department	Average of Salary
Finance	66169.80
Hr	67058.54
It	66722.28
Marketing	66600.70
Sales	69153.39
Unknown	74167.93
Grand Total	68269.25

Country	Count of Employee ID
Australia	5
Germany	9
India	15
UK	18
Unknown	17
USA	18
Grand Total	98

Gender	Sum of Performance Rating
Female	40
Male	46
Not specified	74
Other	47
Unknown	11
Grand Total	218

Pivot Tables were created to summarize and analyze employee data efficiently. Average Salary by Department, Employee Count by Country, and Average Performance Rating by Gender were calculated using PivotTables. This helped in identifying patterns and comparing data categories easily.

Day 16 – Pivot Tables (Part 2)



1. Descriptive Analysis

Descriptive analysis explains what happened in the dataset. Using PivotTables, the employee count by year and average salary by department were identified. The analysis showed that employee count decreased over the years and some departments had higher average salaries than others.

2. Diagnostic Analysis

Diagnostic analysis explains why something happened. From the PivotTable insights, it was observed that employee hiring reduced in later years, which caused the downward trend in employee count. Salary differences between departments may indicate variation in job roles and responsibilities.

3. Predictive Analysis

Predictive analysis helps forecast future trends based on historical data. Using a line chart with a trendline, the employee growth pattern was analyzed. The downward trendline predicts that employee count may continue to decrease in future years if the same trend continues.

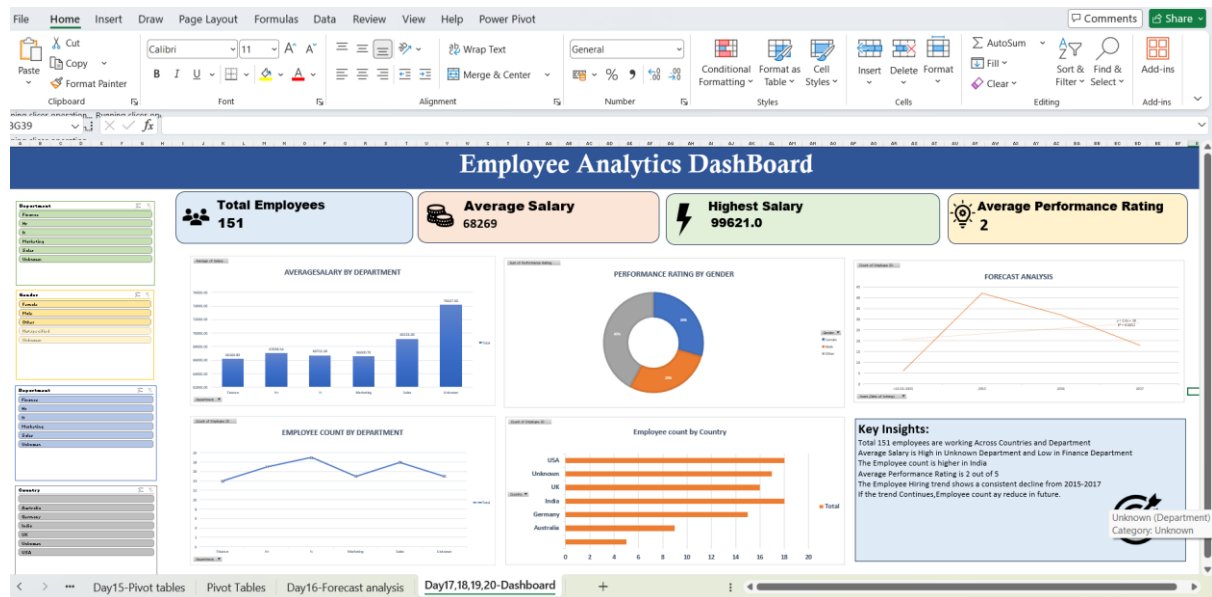
4. Prescriptive Analysis

Prescriptive analysis suggests actions to improve future outcomes. Based on the insights, the company may improve hiring strategies, employee retention, and departmental performance to prevent further decline in employee count and maintain organizational growth.

Day 17-20-Pivot table and Dashboard Creation

A one-page interactive dashboard was created using PivotTables, PivotCharts, KPI cards, slicers, and trend analysis. The dashboard summarized employee distribution, salary analysis, performance insights, and hiring trends. Quick Analysis and AI-assisted Excel tools were explored to generate charts, formulas, and insights efficiently. The dashboard was designed with interactive filters and professional formatting to improve data visualization and storytelling.

Final Dashboard:



- PivotCharts were created directly from PivotTables using: Insert → PivotChart
- The PivotChart automatically linked with the PivotTable data.
- Any changes made in the PivotTable were dynamically reflected in the PivotChart.
- Different chart types such as Column, Pie, Line, and Combo charts were used for better visualization of insights.
- Departments with higher average salaries were identified.
- Employee distribution across countries and genders was analyzed.
- Hiring trends over different years were visualized using line charts and trendlines.
- Performance ratings helped identify top-performing employee groups.
- Missing and inconsistent values were cleaned to improve data quality and reporting accuracy.
- KPI cards were created to display important metrics such as:
 - Total Employees
 - Average Salary
 - Highest Salary
 - Average Performance Rating
- Interactive slicers were added for filtering data by Department, Gender, and Country.
- Charts and PivotTables were arranged systematically to improve storytelling and readability.
- Consistent formatting, colors, titles, and alignment were used to make the dashboard professional and presentation-ready.

Formulas Used:

COUNT() Count numerical values

COUNTA() Count non-empty cells

COUNTIF() Count based on condition

SUMIF() Calculate sum based on condition

AVERAGEIF() Calculate average based on condition

IF() Apply logical conditions

LEFT() Extract characters from left

TRIM() Remove extra spaces

PROPER() Convert text into proper case

YEAR() Extract year from joining date

INDEX() + MATCH() Lookup and retrieve values

DATEDIF() Calculate date differences

MAX() Find highest value

MIN() Find lowest value

